CENGAGE



FOUNDATIONS OF COMPUTER SCIENCE

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FOURTH EDITION

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CHAPTER 6

Computer Networks and Internet



The development of the personal computer has brought about tremendous changes for business, industry, science, and education. A similar revolution has occurred in networking. Technological advances are making it possible for communication links to carry more and faster signals. As a result, services are evolving to allow use of this expanded capacity. Research in this area has resulted in new technologies. One goal is to be able to exchange data such as text, audio, and video from all parts of the world. We want to access the Internet to download and upload information quickly and accurately and at any time.

Objectives

After studying this chapter, the student should be able to:

- Describe local and wide area networks (LANs and WANs).
- Distinguish an internet from the Internet.
- Describe the TCP/IP protocol suite as the network model in the Internet.
- Define the layers in the TCP/IP protocol suite and their relationship.
- Describe some applications at the application layer.
- Describe the services provided by the transport-layer protocols.
- Describe the services provided by the network-layer protocols
- Describe different protocols used at the data-link layer.
- Describe the duties of the physical layer.
- Describe the different transmission media used in computer networking.

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6.1 OVERVIEW

Although the goal of this chapter is to discuss the Internet, a system that interconnects billions of computers in the world, we think of the Internet not as a single network, but as an **internetwork**, a combination of networks. Therefore, we start our journey by first defining a network. We then show how we can connect networks to create small internetworks. Finally, we show the structure of the Internet and open the gate to study the Internet in the rest of this chapter.

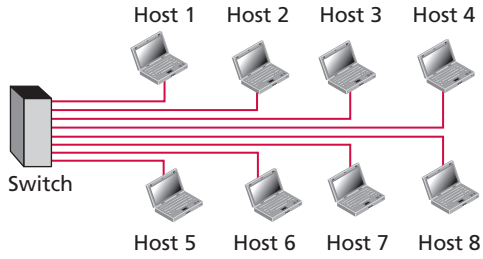
6.1.1 Networks

A **network** is the interconnection of a set of devices capable of communication. In this definition, a device can be a **host** (or an **end system** as it is sometimes called) such as a large computer, desktop, laptop, workstation, cellular phone, or security system. A device in this definition can also be a **connecting device** such as a **router** which connects the network to other networks, a **switch** which connects devices together, a modem (modulator-demodulator) that changes the form of data, and so on. These devices in a network are connected using wired or wireless transmission media such as cable or air. When we connect two computers at home using a plug-and-play router, we have created a network, although very small.

Local area network

A **local area network** (LAN) is usually privately owned and connects some hosts in a single office, building, or campus. Depending on the needs of an organization, a LAN can be as simple as two PCs and a printer in someone's home office, or it can extend throughout a company and include audio and video devices. Each host in a LAN has an identifier, an address, that uniquely defines the host in the LAN. A packet sent by a host to another host carries both the source host's and the destination host's addresses. Figure 6.1 shows an example of a LAN.

Figure 6.1 *Example of a LAN*



Wide area network

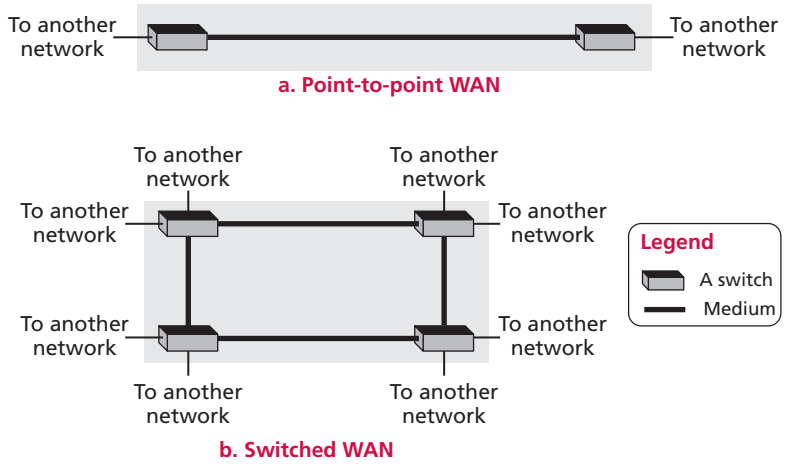
A **wide area network (WAN)** is also an interconnection of devices capable of communication. However, there are some differences between a LAN and a WAN. A LAN is normally

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limited in size, spanning an office, a building, or a campus; a WAN has a wider geographical span, spanning a town, a state, a country, or even the world. A LAN interconnects hosts; a WAN interconnects connecting devices such as switches, routers, or modems. A LAN is normally privately owned by the organization that uses it; a WAN is normally created and run by communication companies and leased by an organization that uses it. We see two distinct examples of WANs today: **point-to-point** WANs and **switched WANs** as shown in Figure 6.2.

Figure 6.2 *A point-to-point and a switched WAN*



A point-to-point WAN is a network that connects two communicating devices through a transmission medium (cable or air).

A switched WAN is a network with more than two ends. A switched WAN, as we will see shortly, is used in the backbone of global communication today. We can say that a switched WAN is a combination of several point-to-point WANs that are connected by switches.

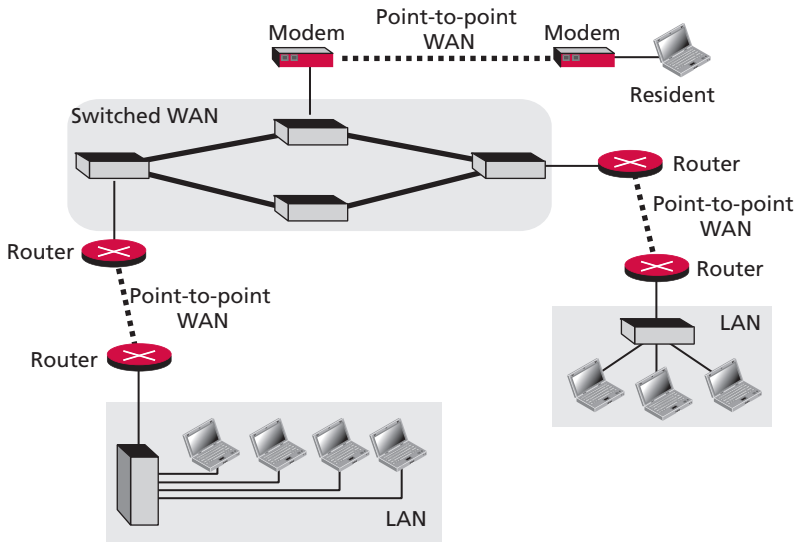
Internetwork

Today, it is very rare to see a LAN or a WAN in isolation; they are connected to one another. When two or more networks are connected, they make an internetwork, or **internet**. As an example, assume that an organization has two offices. Each office has a LAN that allows all employees in the office to communicate with each other. To make the communication between employees at different offices possible, the management leases a point-to-point dedicated WAN from a service provider, such as a telephone company, and connects the two LANs. Now the company has an internetwork, or a private internet (with lowercase *i*). Communication between offices is now possible. Figure 6.3 shows this internet.

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Figure 6.3 *An internetwork made of two LANs and three WANs*



6.1.2 The Internet

As we discussed before, an internet (note the lowercase *i*) is two or more networks that can communicate with each other. The most notable internet is called the **Internet** (uppercase *I*), and is composed of thousands of interconnected networks. Figure 6.4 shows a conceptual (not geographical) view of the Internet.

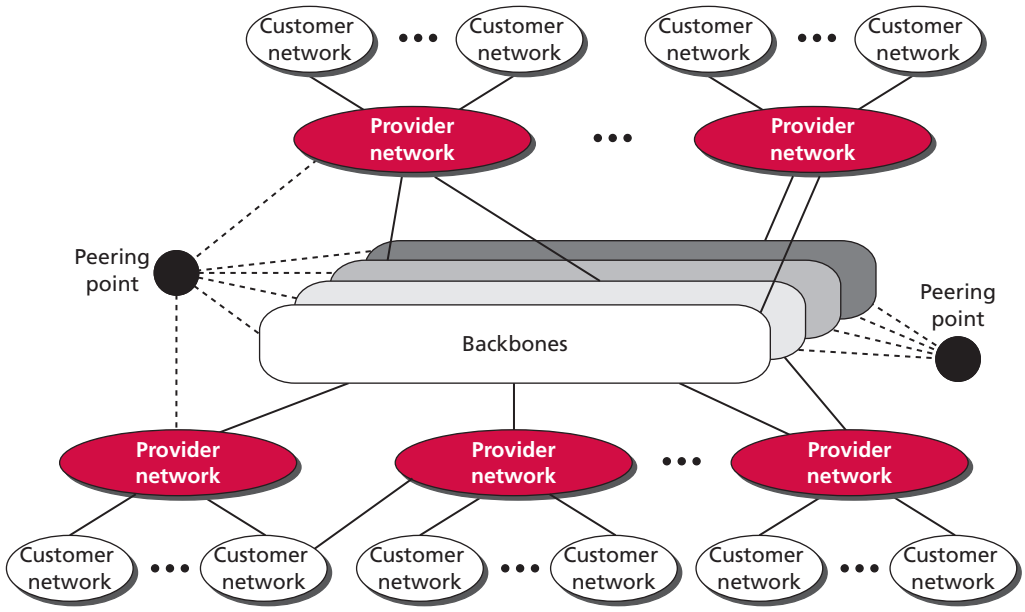
The figure shows the Internet as several backbones, provider networks, and customer networks. At the top level, the *backbones* are large networks owned by some communication companies. The backbone networks are connected through some complex switching systems, called *peering points*. At the second level, there are smaller networks, called *provider networks*, that use the services of the backbones for a fee. The provider networks are connected to backbones and sometimes to other provider networks. The *customer networks* are networks at the edge of the Internet that actually use the services provided by the Internet. They pay fees to provider networks for receiving services.

Backbones and provider networks are also called **internet service providers (ISPs)**. The backbones are often referred to as international ISPs; the provider networks are often referred to as national or regional ISPs.

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Figure 6.4 *The Internet today*



6.1.3 Hardware and software

We have given the overview of the Internet structure, which is made of small and large networks glued together with connecting devices. It should be clear, however, that if we only connect these pieces nothing will happen. For communication to happen, we need both **hardware** and **software**. This is similar to a complex computation in which we need both a computer and a program. In the next section, we show how these combinations of hardware and software are coordinated with each other using *protocol layering*.

6.1.4 Protocol layering

A word we hear all the time when we talk about the Internet is *protocol*. A **protocol** defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively. When communication is simple, we may need only one simple protocol; when the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or **protocol layering**.

A scenario

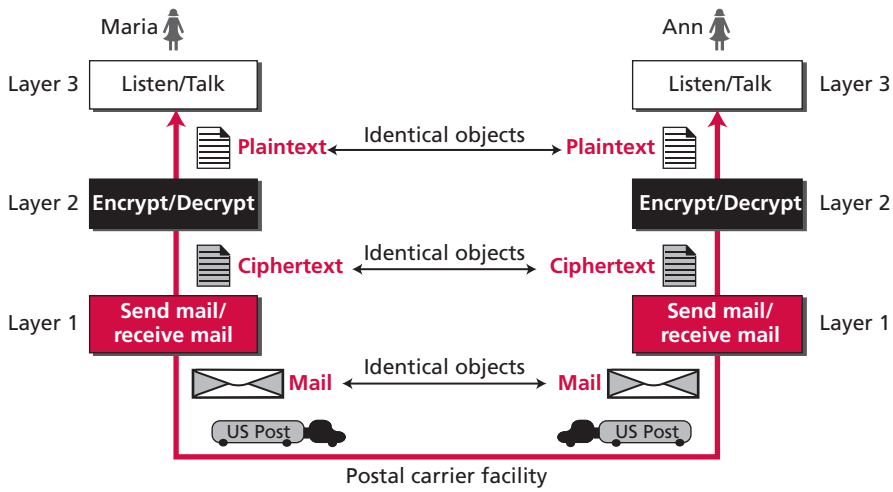
Let us develop a simple scenario to better understand the need for protocol layering. We assume that Ann and Maria are neighbors with a lot of common ideas. They communicate with each other every time they meet about a project for the time when they will be retired.

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Suddenly, Ann is offered a higher-level position in her company, but needs to move to another branch located in a city very far from Maria. The two friends still want to continue their communication and exchange ideas because they have come up with an innovative project to start a new business when they both retire. They decide to continue their conversation using regular mail through the post office. However, they do not want their ideas to be revealed to other people if the letters are intercepted. They agree on an encryption/decryption technique. The sender of the letter encrypts it to make it unreadable by an intruder; the receiver of the letter decrypts it to get the original letter. We discuss the encryption/decryption methods in Chapter 16, but for the moment we assume that Maria and Ann use one technique that makes it hard to decrypt the letter if one does not have the key for doing so. Now we can say that the communication between Maria and Ann takes place in three layers, as shown in Figure 6.5. We assume that Ann and Maria each have three machines (or robots) that can perform the task at each layer.

Figure 6.5 *A three-layer protocol*



Let us assume that Maria sends the first letter to Ann. Maria talks to the machine at the third layer as though the machine is Ann and is listening to her. The third-layer machine listens to what Maria says and creates the plaintext (a letter in English), which is passed to the second-layer machine. The second-layer machine takes the plaintext, encrypts it, and creates the **ciphertext**, which is passed to the first-layer machine. The first layer machine, presumably a robot, takes the ciphertext, puts it in an envelope, adds the sender and receiver addresses, and mails it.

At Ann's side, the first-layer machine picks up the letter from Ann's mail box, recognizing the letter from Maria by the sender address. The machine takes out the ciphertext from the envelope and delivers it to the second-layer machine. The second-layer machine decrypts the message, creates the plaintext, and passes the plaintext to the third-layer machine. The third-layer machine takes the plaintext and reads it as though Maria is speaking.

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Protocol layering enables us to divide a complex task into several smaller and simpler tasks. For example, in Figure 6.5, we could have used only one machine to do the job of all three machines. However, if Maria and Ann decide that the encryption/decryption done by the machine is not enough to protect their secrecy, they have to change the whole machine. In the present situation, they need to change only the second-layer machine; the other two can remain the same. This is referred to as **modularity**. Modularity in this case means independent layers. A layer **(module)** can be defined as a black box with inputs and outputs, without concern about how inputs are changed to outputs. If two machines provide the same outputs when given the same inputs, they can replace each other. For example, Ann and Maria can buy the second-layer machine from two different manufacturers. As long as the two machines create the same ciphertext from the same plaintext and *vice versa*, they do the job.

One of the advantages of protocol layering is that it allows us to separate the services from the implementation. A layer needs to be able to receive a set of services from the lower layer and to give the services to the upper layer; we don't care about how the layer is implemented. For example, Maria may decide not to buy the machine (robot) for the first layer; she can do the job herself. As long as Maria can do the tasks provided by the first layer, in both directions, the communication system works.

Another advantage of protocol layering, which cannot be seen in our simple examples, but reveals itself when we discuss protocol layering in the Internet, is that communication does not always use only two end systems; there are intermediate systems that need only some layers, but not all layers. If we did not use protocol layering, we would have to make each intermediate system as complex as the end systems, which makes the whole system more expensive.

Is there any disadvantage to protocol layering? One can argue that having a single layer makes the job easier. There is no need for each layer to provide a service to the upper layer and give a service to the lower layer. For example, Ann and Maria could find or build one machine that could do all three tasks. However, as mentioned above, if one day they found that their code was broken, each would have to replace the whole machine with a new one instead of just changing the machine in the second layer.

Principles of protocol layering

Let us discuss some principles of protocol layering. The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction. For example, the third-layer task is to *listen* (in one direction) and *talk* (in the other direction). The second layer needs to be able to encrypt and decrypt. The first layer needs to send and receive mail.

The second important principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical. For example, the object under layer 3 at both sites should be a plaintext letter. The object under layer 2 at both sites should be a ciphertext letter. The object under layer 1 at both sites should be a piece of mail.

Logical connections

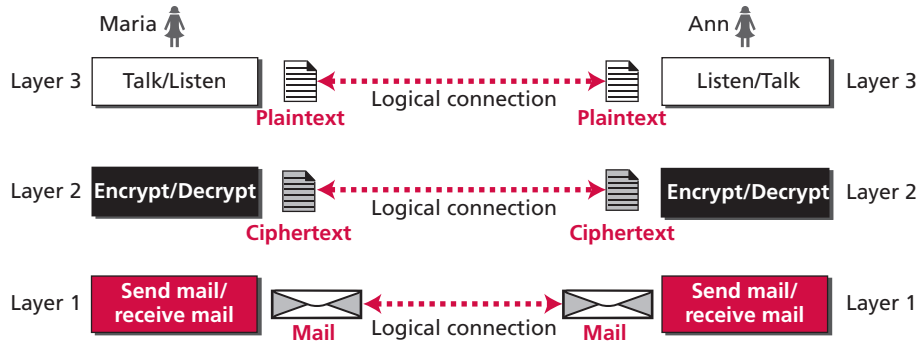
After following the above two principles, we can think about logical connection between each layer as shown in Figure 6.6. This means that we have layer-to-layer communication.

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Maria and Ann can think that there is a logical (imaginary) connection at each layer through which they can send the object created from that layer. We will see that the concept of logical connection will help us better understand the task of layering we encounter in data communication and networking.

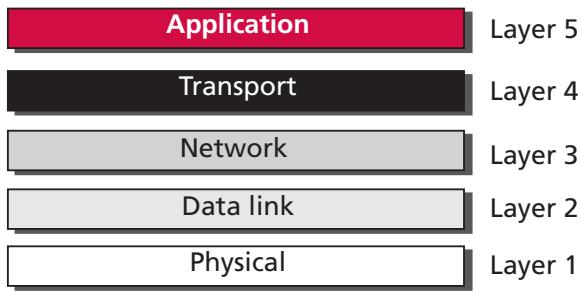
Figure 6.6 *Logical connection between peer layers*



6.1.5 TCP/IP protocol suite

Now that we know about the concept of protocol layering and the logical communication between layers in our second scenario, we can introduce the **TCP/IP (Transmission Control Protocol/Internet Protocol)**. TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today. It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality. The term *hierarchical* means that each upper-level protocol is supported by the services provided by one or more lower-level protocols. The TCP/IP protocol suite is made of five layers as shown in Figure 6.7.

Figure 6.7 *Layers in the TCP/IP protocol suite*



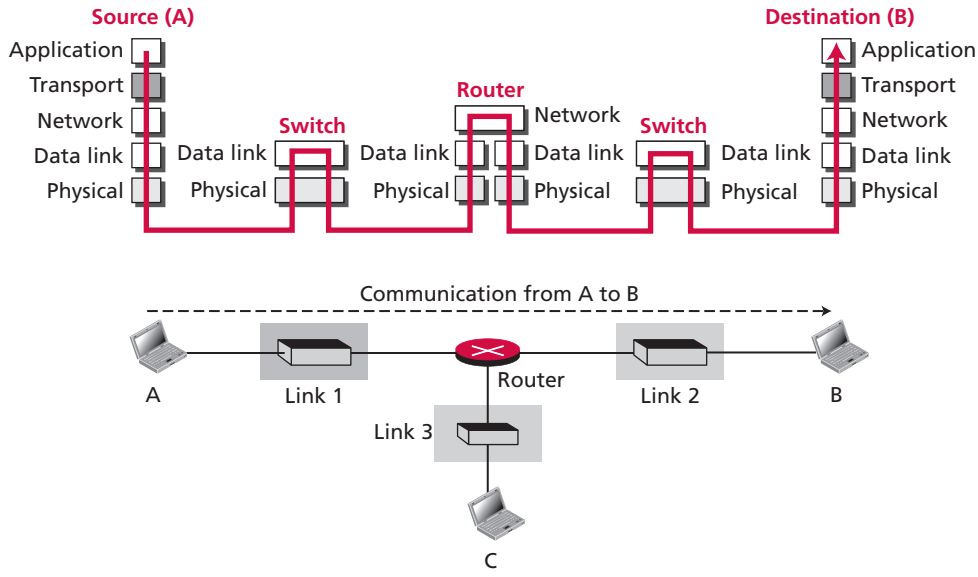
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Layered architecture

To show how the layers in the TCP/IP protocol suite are involved in communication between two hosts, we assume that we want to use the suite in a small internet made up of three LANs (links), each with a link-layer switch. We also assume that the links are connected by one router, as shown in Figure 6.8.

Figure 6.8 *Communication through an internet*



Let us assume that computer A communicates with computer B. As the figure shows, we have five communicating devices in this communication: source host (computer A), the link-layer switch in link 1, the router, the link-layer switch in link 2, and the destination host (computer B). Each device is involved with a set of layers depending on the role of the device in the internet. The two hosts are involved in all five layers; the source host needs to create a message in the **application layer** and send it down the layers so that it is physically sent to the destination host. The destination host needs to receive the communication at the **physical layer** and then deliver it through the other layers to the application layer.

The router is involved only in three layers; there is no transport or application layer in a router as long as the router is used only for **routing**. Although a router is always involved in one network layer, it is involved in *n* combinations of link and physical layers in which *n* is the number of links the router is connected to. The reason is that each link may use its own data-link or physical protocol. For example, in the above figure, the router is involved in three links, but the message sent from source A to destination B is involved in two links. Each link may be using different link-layer and physical-layer protocols; the router needs to receive a packet from link 1 based on one pair of protocols and deliver it to link 2 based on another pair of protocols.

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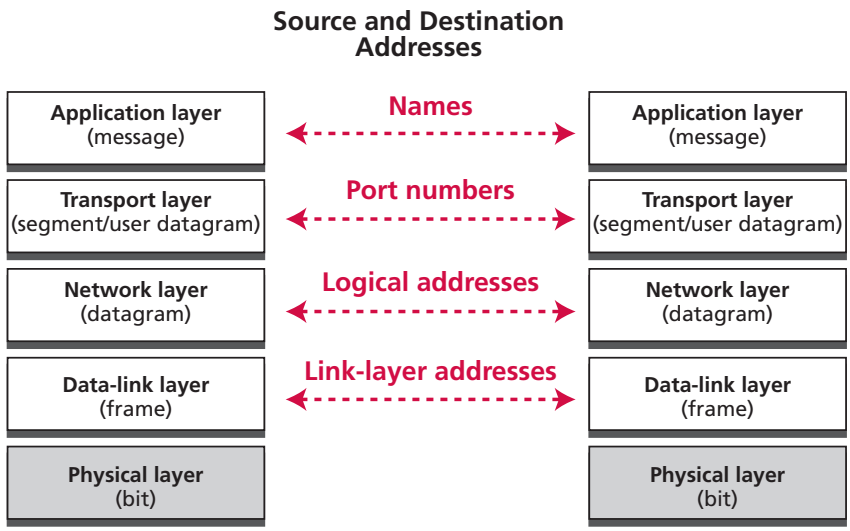
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A link-layer switch in a link, however, is involved only in two layers, data-link and physical. Although each switch in the above figure has two different connections, the connections are in the same link, which uses only one set of protocols. This means that, unlike a router, a link-layer switch is involved only in one data-link and one physical layer.

Addressing and packet names

It is worth mentioning another two concepts related to protocol layering in the Internet: *addressing* and *packet names*. As we discussed before, we have logical communication between pairs of layers in this model. Any communication that involves two parties needs two addresses: source address and destination address. Although it looks as if we need five pairs of addresses, one pair per layer, we normally have only four because the physical layer does not need addresses; the unit of data exchange at the physical layer is a bit, which definitely cannot have an address. Figure 6.9 shows the addressing and the names of the packets at each layer.

Figure 6.9 *Addressing and packet names in the TCP/IP protocol suite*



As the figure shows, there is a relationship between the layer, the address used in that layer, and the packet name at that layer. At the application layer, we normally use names to define the site that provides services, such as *someorg.com*, or the email address, such as *somebody@coldmail.com*. At the transport layer, addresses are called **port numbers**, and these define the application-layer programs at the source and destination. Port numbers are local addresses that distinguish between several programs running at the same time. At the **network-layer**, the addresses are global, with the whole Internet as the scope. A network-layer address uniquely defines the connection of a device to the Internet. The link-layer

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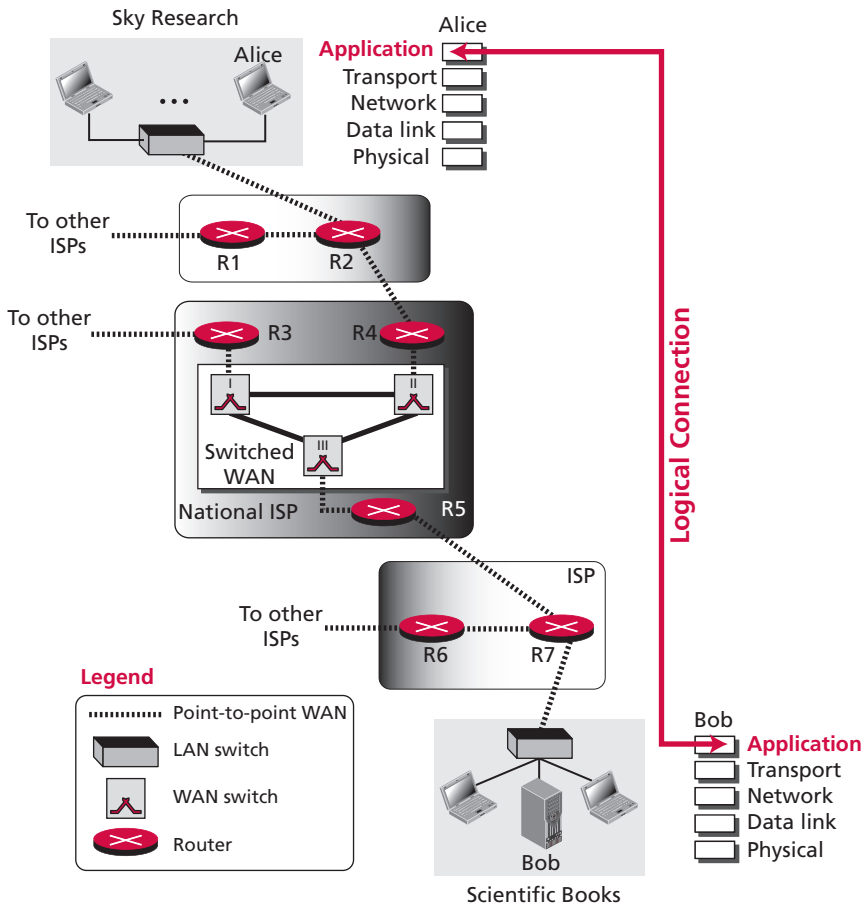
addresses, sometimes called **MAC addresses**, are locally defined addresses, each of which defines a specific host or router in a network (LAN or WAN).

6.2 APPLICATION LAYER

After the brief discussion of networks, internetworks, and the Internet, we are ready to give some discussion about each layer of the TCP/IP protocol. We start from the fifth layer and move to the first layer.

The fifth layer of the TCP/IP protocol is called the application layer. The application layer provides services to the user. Communication is provided using a logical connection, which means that the two application layers assume that there is an imaginary direct connection through which they can send and receive messages. Figure 6.10 shows the idea behind this logical connection.

Figure 6.10 *Logical connection at the application layer*



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The figure shows a scenario in which a scientist working in a research company, Sky Research, needs to order a book related to her research from an online bookseller, Scientific Books. Logical connection takes place between the application layer at a computer at Sky Research and the application layer of a **server** at Scientific Books. We call the first host Alice and the second one Bob. The communication at the application layer is logical, not physical. Alice and Bob assume that there is a two-way logical channel between them through which they can send and receive messages. The actual communication, however, takes place through several devices (Alice, R2, R4, R5, R7, and Bob) and several physical channels as shown in the figure.

6.2.1 Providing services

The application layer is somehow different from other layers in that it is the highest layer in the suite. The protocols in this layer do not provide services to any other protocol in the suite; they only receive services from the protocols in the transport layer. This means that protocols can be removed from this layer easily. New protocols can be also added to this layer as long as the new protocol can use the service provided by one of the transport-layer protocols.

Since the application layer is the only layer that provides services to the Internet user, the flexibility of the application layer, as described above, allows new application protocols to be easily added to the Internet, which has been occurring during the lifetime of the Internet. When the Internet was created, only a few application protocols were available to the users; today we cannot give a number for these protocols because new ones are being added constantly.

6.2.2 Application-layer paradigms

It should be clear that to use the Internet we need two application programs to interact with each other: one running on a computer somewhere in the world, the other running on another computer somewhere else in the world. The two programs need to send messages to each other through the Internet infrastructure. However, we have not discussed what the relationship should be between these programs. Should both application programs be able to request services and provide services, or should the application programs just do one or the other? Two paradigms have been developed during the lifetime of the Internet to answer this question: the *client–server paradigm* and the *peer-to-peer paradigm*. We briefly introduce these two paradigms here.

Traditional paradigm: client–server

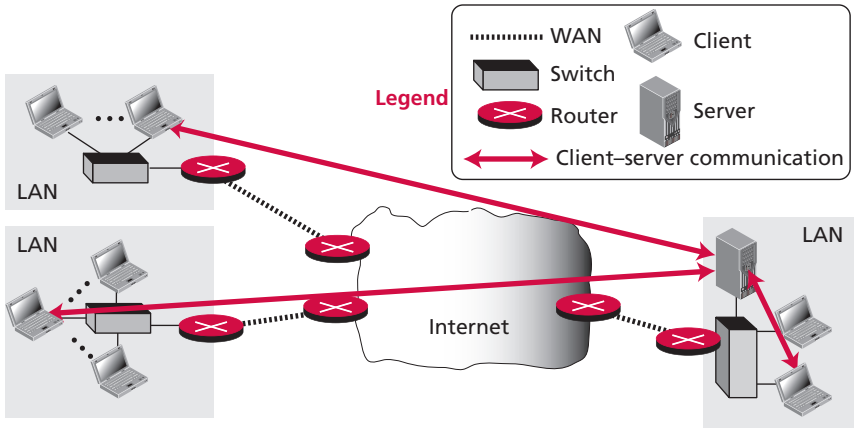
The traditional paradigm is called the **client–server paradigm**. It was the most popular paradigm until a few years ago. In this paradigm, the service provider is an application program, called the server process; it runs continuously, waiting for another application program, called the client process, to make a connection through the Internet and ask for service. There are normally some server processes that can provide a specific type of service, but there are many clients that request service from any of these server processes. The server process must be running all the time; the client process is started when the client needs to receive service.

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Although the communication in the client–server paradigm is between two application programs, the role of each program is totally different. In other words, we cannot run a client program as a server program or *vice versa*. Figure 6.11 shows an example of a client–server communication in which three clients communicate with one server to receive the services provided by this server.

Figure 6.11 *Example of a client–server paradigm*



One problem with this paradigm is that the concentration of the communication load is on the shoulder of the server, which means the server should be a powerful computer. Even a powerful computer may become overwhelmed if a large number of clients try to connect to the server at the same time. Another problem is that there should be a service provider willing to accept the cost and create a powerful server for a specific service, which means the service must always return some type of income for the server in order to encourage such an arrangement.

Several traditional services are still using this paradigm, including the **World Wide Web (WWW)** and its vehicle HyperText Transfer Protocol (HTTP), file transfer protocol (FTP), secure shell (SSH), email, and so on. We discuss some of these protocols and applications later in the chapter.

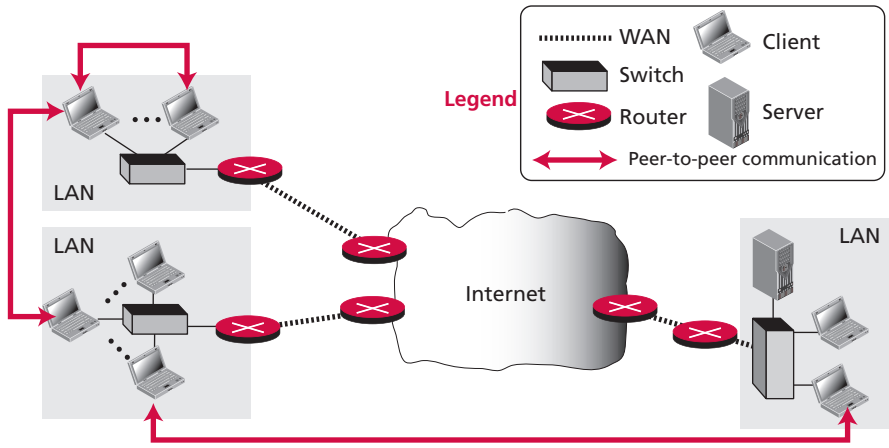
New paradigm: peer-to-peer

A new paradigm, called the **peer-to-peer paradigm** (often abbreviated *P2P paradigm*) has emerged to respond to the needs of some new applications. In this paradigm, there is no need for a server process to be running all the time and waiting for the client processes to connect. The responsibility is shared between peers. A computer connected to the Internet can provide service at one time and receive service at another time. A computer can even provide and receive services at the same time. Figure 6.12 shows an example of communication in this paradigm.

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Figure 6.12 *Example of a peer-to-peer paradigm*



One of the areas that really fits in this paradigm is Internet telephony. Communication by phone is indeed a peer-to-peer activity; no party needs to be running forever waiting for the other party to call. Another area in which the peer-to-peer paradigm can be used is when some computers connected to the Internet have something to share with each other. For example, if an Internet user has a file available to share with other Internet users, there is no need for the file holder to become a server and run a server process all the time waiting for other users to connect and retrieve the file.

Although the peer-to-peer paradigm has been proved to be easily scalable and cost-effective in eliminating the need for expensive servers to be running and maintained all the time, there are also some challenges. The main challenge has been **security**; it is more difficult to create secure communication between distributed services than between those controlled by some dedicated servers. The other challenge is applicability; it appears that not all applications can use this new paradigm. For example, not many Internet users are ready to become involved, if one day the Web can be implemented as a peer-to-peer service.

6.2.3 Standard client–server applications

During the lifetime of the Internet, several client–server application programs have been developed. We do not have to redefine them, but we need to understand what they do.

We have selected six standard application programs in this section. We start with HTTP and the World Wide Web because they are used by almost all Internet users. We then introduce file transfer and electronic mail applications which have high traffic loads on the Internet. Next, we explain remote logging and how it can be achieved using the TELNET and SSH protocols. Finally, we discuss DNS, which is used by all application programs to map the application layer identifier to the corresponding host **IP address**.

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World Wide Web and HTTP

In this section, we first introduce the World Wide Web (abbreviated WWW or Web). We then discuss the HyperText Transfer Protocol (HTTP), the most common client–server application program used in relation to the Web.

World Wide Web

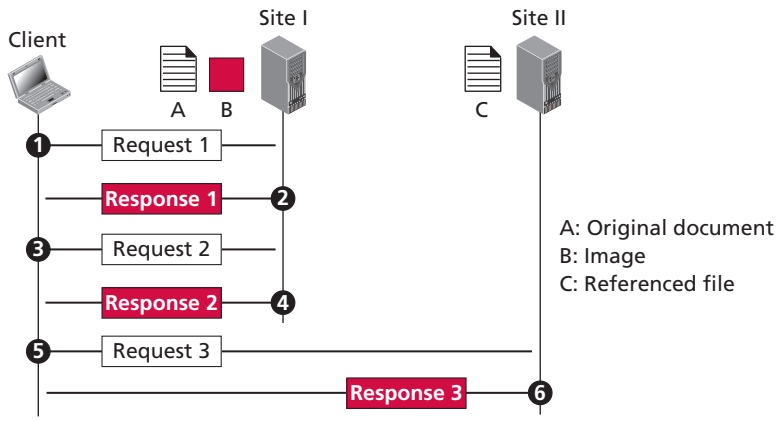
The Web today is a repository of information in which the documents, called Web pages, are distributed all over the world and related documents are linked together. The popularity and growth of the Web can be related to two terms in the above statement: distributed and linked. Distribution allows the growth of the Web. Each web server in the world can add a new web page to the repository and announce it to all Internet users without overloading a few servers. Linking allows one web page to refer to another web page stored in another server somewhere else in the world. The linking of web pages was achieved using a concept called **hypertext**, which was introduced many years before the advent of the Internet. The idea was to use a machine that automatically retrieved another document stored in the system when a link to it appeared in the document. The Web implemented this idea electronically: to allow the linked document to be retrieved when the link was clicked by the user. Today, the term hypertext, coined to mean linked text documents, has been changed to hypermedia, to show that a web page can be a text document, an image, an audio file, or a video file.

The WWW today is a distributed client–server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called *sites*. Each site holds one or more documents, referred to as web pages. Each **web page**, however, can contain some links to other web pages in the same or other sites. In other words, a web page can be simple or composite. A simple web page has no links to other web pages; a composite web page has one or more links to other web pages. Each web page is a file with a name and address.

Example 6.1

Assume we need to retrieve a scientific document that contains one reference to another text file and one reference to a large image. Figure 6.13 shows the situation.

Figure 6.13 *Example 6.1*



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The main document and the image are stored in two separate files in the same site (file A and file B); the referenced text file is stored in another site (file C). Since we are dealing with three different files, we need three transactions if we want to see the whole document. The first transaction (request/response) retrieves a copy of the main document (file A), which has references (pointers) to the second and third files. When a copy of the main document is retrieved and browsed, the user can click on the reference to the image to invoke the second transaction and retrieve a copy of the image (file B). If the user needs to see the contents of the referenced text file, he or she can click on its reference (pointer) invoking the third transaction and retrieving a copy of file C. Note that although files A and B both are stored in site I, they are independent files with different names and addresses. Two transactions are needed to retrieve them. A very important point we need to remember is that file A, file B, and file C in Example 6.1 are independent web pages, each with independent names and addresses. Although references to file B or C are included in file A, it does not mean that each of these files cannot be retrieved independently. A second user can retrieve file B with one transaction. A third user can retrieve file C with one transaction.

Web client (browser)

A variety of vendors offer commercial **browsers** that interpret and display a web page, and all of them use nearly the same architecture. Each browser usually consists of three parts: a controller, client protocols, and interpreters.

The controller receives input from the keyboard or the mouse and uses the client programs to access the document. After the document has been accessed, the controller uses one of the interpreters to display the document on the screen. The client protocol can be one of the protocols described later, such as HTTP or FTP. The interpreter can be **HyperText Markup Language (HTML)**, Java, or JavaScript, depending on the type of document. Some commercial browsers include Internet Explorer, Netscape Navigator, and Firefox.

Web server

The web page is stored at the server. Each time a request arrives, the corresponding document is sent to the client.

Uniform resource locator (URL)

A web page, as a file, needs to have a unique identifier to distinguish it from other web pages. To define a web page, we need three identifiers: *host*, *port*, and *path*. However, before defining the web page, we need to tell the browser what client–server application we want to use, which is called the *protocol*. This means we need four identifiers to define the web page. The first is the type of vehicle to be used to fetch the web page; the last three make up the combination that defines the destination object (web page).

- **Protocol**. The first identifier is the abbreviation for the client–server program that we need in order to access the web page.
- **Host identifier**. The host identifier can be the IP address of the server or the unique name given to the server.
- **Port number**. The port, a 16-bit integer, is normally predefined for the client–server application.
- **Path**. The path identifies the location and the name of the file in the underlying operating system. The format of this identifier normally depends on the operating system.

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In UNIX, a path is a set of directory names followed by the file name, all separated by a slash.

To combine these four pieces together, the **uniform resource locator (URL)** has been designed; it uses three different separators between the four pieces as shown below:

protocol://host/path	**Used most of the time**
protocol://host:port/path	**Used when port number is needed**

HyperText Transfer Protocol (HTTP)

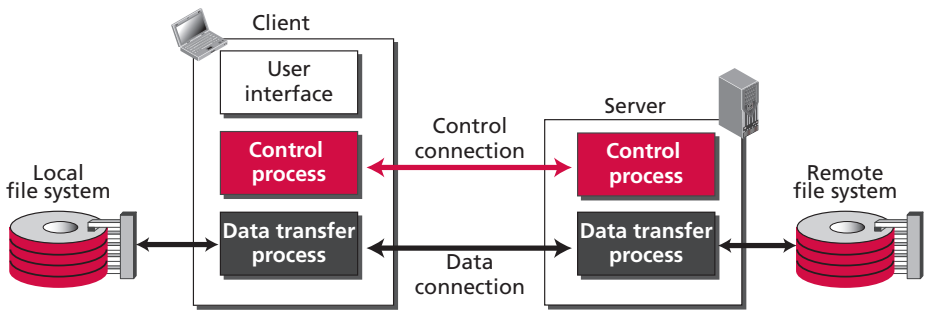
The **HyperText Transfer Protocol (HTTP)** is a protocol that is used to define how the client–server programs can be written to retrieve web pages from the Web. An HTTP client sends a request; an HTTP server returns a response. The server uses the port number 80; the client uses a temporary port number.

6.2.4 File Transfer Protocol (FTP)

File Transfer Protocol (FTP) is the standard protocol provided by TCP/IP for copying a file from one host to another. Although transferring files from one system to another seems simple and straightforward, some problems must be dealt with first. For example, two systems may use different file name conventions. Two systems may have different ways to represent data. Two systems may have different directory structures. All of these problems have been solved by FTP in a very simple and elegant approach.

Figure 6.14 shows the basic model of FTP. The client has three components: user interface, client control process, and the client data transfer process. The server has two components: the server control process and the server data transfer process. The control connection is made between the control processes. The data connection is made between the data transfer processes.

Figure 6.14 *FTP*



Separation of commands and data transfer makes FTP more efficient. The control connection uses very simple rules of communication. We need to transfer only a line of command or a line of response at a time. The data connection, on the other hand, needs more complex rules due to the variety of data types transferred.

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Lifetimes of two connections

The two connections in FTP have different lifetimes. The control connection remains connected during the entire interactive FTP session. The data connection is opened and then closed for each file transfer activity. It opens each time commands that involve transferring files are used, and it closes when the file is transferred. In other words, when a user starts an FTP session, the control connection opens. While the control connection is open, the data connection can be opened and closed multiple times if several files are transferred.

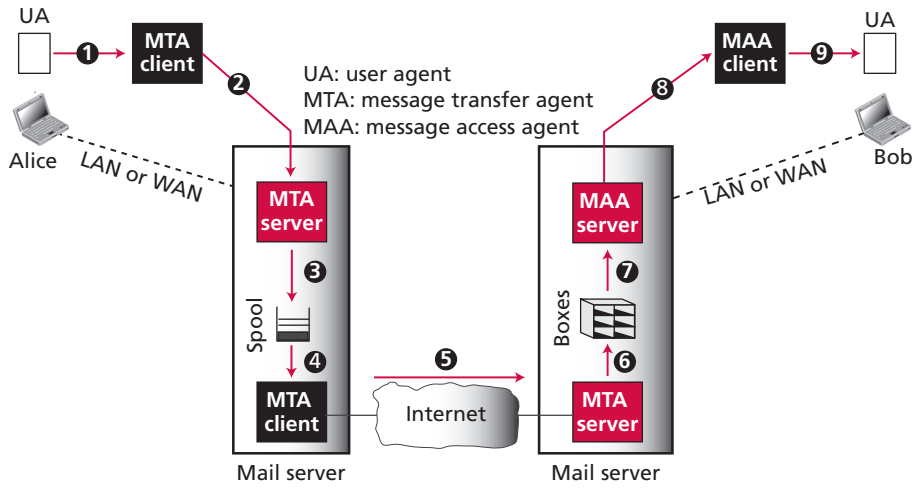
6.2.5 Electronic mail

Electronic mail (or email) allows users to exchange messages. The nature of this application, however, is different from other applications discussed so far. In an application such as HTTP or FTP, the server program is running all the time, waiting for a request from a client. When the request arrives, the server provides the service. There is a request and there is a response. In the case of electronic mail, the situation is different. First, email is considered a one-way transaction. When Alice sends an email to Bob, she may expect a response, but this is not a mandate. Bob may or may not respond. If he does respond, it is another one-way transaction. Second, it is neither feasible nor logical for Bob to run a server program and wait until someone sends an email to him. Bob may turn off his computer when he is not using it. This means that the idea of client/server programming should be implemented in another way: using some intermediate computers (servers). The users run only client programs when they want and the intermediate servers apply the client/server paradigm, as we discuss in the next section.

Architecture

To explain the architecture of email, we give a common scenario, as shown in Figure 6.15.

Figure 6.15 *Common scenario*



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In the common scenario, the sender and the receiver of the email, Alice and Bob respectively, are connected via a LAN or a WAN to two mail servers. The administrator has created one mailbox for each user where the received messages are stored. A *mailbox* is part of a server hard drive, a special file with permission restrictions. Only the owner of the mailbox has access to it. The administrator has also created a queue (spool) to store messages waiting to be sent.

A simple email from Alice to Bob takes nine different steps, as shown in the figure. Alice and Bob use three different *agents*: a **user agent (UA)**, a **Mail Transfer Agent (MTA)**, and a **Message Access Agent (MAA)**. When Alice needs to send a message to Bob, she runs a UA program to prepare the message and send it to her mail server. The mail server at her site uses a queue (spool) to store messages waiting to be sent. The message, however, needs to be sent through the Internet from Alice's site to Bob's site using an MTA. Here two **message transfer agents** are needed: one client and one server. Like most client–server programs on the Internet, the server needs to run all the time because it does not know when a client will ask for a connection. The client, on the other hand, can be triggered by the system when there is a message in the queue to be sent. The user agent at Bob's site allows Bob to read the received message. Bob later uses an MAA client to retrieve the message from an MAA server running on the second server.

Bob needs another pair of client–server programs: message access programs. This is because an MTA client–server program is a *push* program: the client pushes the message to the server. Bob needs a *pull* program. The client needs to pull the message from the server. We discuss more about MAAs shortly.

6.2.6 TELNET

A server program can provide a specific service to its corresponding client program. For example, the FTP server is designed to let the FTP client store or retrieve files on the server site. However, it is impossible to have a client/server pair for each type of service we need; the number of servers soon becomes intractable. The idea is not scalable. Another solution is to have a specific client/server program for a set of common scenarios, but to have some generic client/server programs that allow a user on the client site to log into the computer at the server site and use the services available there. For example, if a student needs to use the Java compiler program at her university lab, there is no need for a Java compiler client and a Java compiler server. The student can use a client logging program to log into the university server and use the compiler program at the university. We refer to these generic client/server pairs as **remote login** applications.

One of the original remote logging protocols is **TELNET**, which is an abbreviation for *TErminaL NETwork*. Although TELNET requires a logging name and password, it is vulnerable to hacking because it sends all data including the password in plaintext (not encrypted). A hacker can eavesdrop and obtain the logging name and password. Because of this security issue, the use of TELNET has diminished in favor of another protocol, Secure Shell (SSH), which we describe in the next section.

6.2.7 Secure Shell (SSH)

Although **Secure Shell (SSH)** is a secure application program that can be used today for several purposes such as remote logging and file transfer, it was originally designed to replace

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TELNET. There are two versions of SSH: SSH-1 and SSH-2, which are totally incompatible. The first version, SSH-1, is now deprecated because of security flaws in it. The current version is called SSH-2.

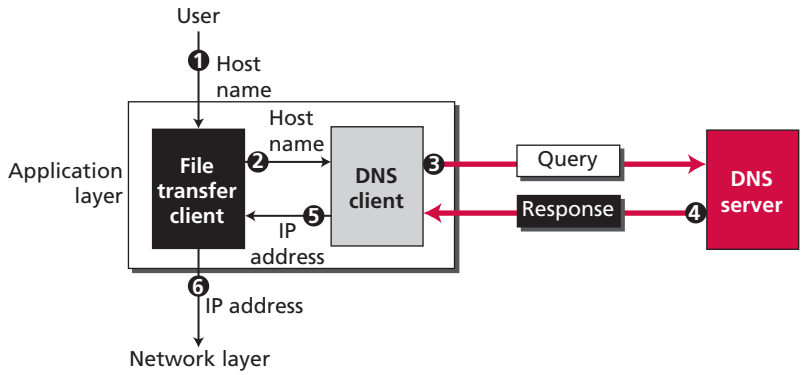
6.2.8 Domain Name System (DNS)

The last client–server application program we discuss has been designed to help other application programs. To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet. However, people prefer to use names instead of numeric addresses. Therefore, the Internet needs to have a directory system that can map a name to an address. This is analogous to the telephone network. A telephone network is designed to use telephone numbers, not names. People can either keep a private file to map a name to the corresponding telephone number or can call the telephone directory to do so.

Since the Internet is so huge today, a central directory system cannot hold all the mapping. In addition, if the central computer fails, the whole communication network will collapse. A better solution is to distribute the information among many computers in the world. In this method, the host that needs mapping can contact the closest computer holding the needed information. Figure 6.16 shows how TCP/IP uses a **DNS** client and a DNS server to map a name to an address. A user wants to use a file transfer client to access the corresponding file transfer server running on a remote host. The user knows only the file transfer server name, such as *afilesource.com*. However, the TCP/IP suite needs the IP address of the file transfer server to make the connection. The following six steps map the host name to an IP address:

1. The user passes the host name to the file transfer client.
2. The file transfer client passes the host name to the DNS client.
3. Each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.
5. The DNS client passes the IP address to the file transfer server.
6. The file transfer client now uses the received IP address to access the file transfer server.

Figure 6.16 *DNS usage*



https://sanet.st/blogs/polatebooks/

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Name space

To be unambiguous, the names assigned to machines must be carefully selected from a name space with complete control over the binding between the names and IP addresses. In other words, the names must be unique because the addresses are unique. A **name space** can map each address to a unique name and is normally organized hierarchically. In a *hierarchical name space*, each name is made of several parts. The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on. In this case, the authority to assign and control the name spaces can be decentralized. A central authority can assign the part of the name that defines the nature of the organization and the name of the organization. The responsibility for the rest of the name can be given to the organization itself. The organization can add suffixes (or prefixes) to the name to define its host or resources. The management of the organization need not worry that the prefix chosen for a host is taken by another organization because, even if part of an address is the same, the whole address is different. For example, assume two organizations call one of their computers *caesar*. The first organization is given a name by the central authority, such as *first.com*, the second organization is given the name *second.com*. When each of these organizations adds the name *caesar* to the name they have already been given, the end result is two distinguishable names: *ceasar.first.com* and *ceasar.second.com*. The names are unique.

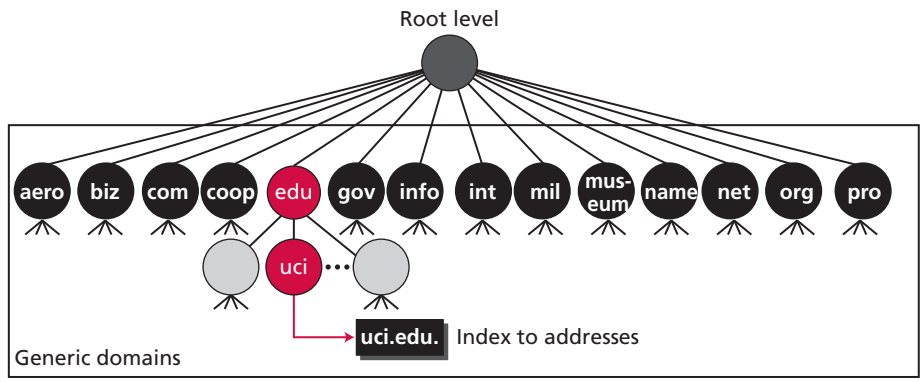
DNS in the Internet

DNS is a protocol that can be used in different platforms. In the Internet, the **domain name space** (tree) was originally divided into three different sections: generic domains, country domains, and the inverse domain. However, due to the rapid growth of the Internet, it became extremely difficult to keep track of the inverse domains, which could be used to find the name of a host when given the IP address. The inverse domains are now deprecated (see RFC 3425). We, therefore, concentrate on the first two.

Generic domains

The **generic domains** define registered hosts according to their generic behavior. Each node in the tree defines a domain, which is an index to the **domain name** space database (see Figure 6.17).

Figure 6.17 *Generic domains*



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Looking at the tree, we see that the first level in the generic domains section allows 14 possible labels. These labels describe the organization types as listed in Table 6.1.

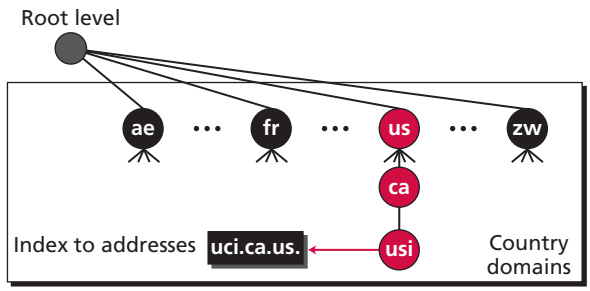
Table 6.1 Generic domain labels

Label	*Description*	*Label*	*Description*
aero	**Airlines and aerospace**	**int**	**International organizations**
biz	**Businesses or firms**	**mil**	**Military groups**
com	**Commercial organizations**	**museum**	**Museums**
coop	**Cooperative organizations**	**name**	**Personal names (individuals)**
edu	**Educational institutions**	**net**	**Network support centers**
gov	**Government institutions**	**org**	**Nonprofit organizations**
info	**Information service providers**	**pro**	**Professional organizations**

Country domains

The **country domains** section uses two-character country abbreviations (e.g., us for United States). Second labels can be organizational, or they can be more specific, national designations. The United States, for example, uses state abbreviations as a subdivision of us (e.g., ca.us.). Figure 6.18 shows the country domains section. The address uci.ca.us. can be translated to University of California, Irvine university in the state of California in the United States.

Figure 6.18 *Country domains*



6.2.9 Peer-to-peer paradigm

We discussed the client–server paradigm early in the chapter. The first instance of peer-to-peer file sharing goes back to December 1987 when Wayne Bell created *WWIVnet*, the network component of WWIV (World War Four) bulletin board software. In July 1999, Ian Clarke designed *Freenet*, a decentralized, censorship-resistant distributed data store, aimed to provide freedom of speech through a peer-to-peer network with strong protection of anonymity.

Peer-to-peer gained popularity with Napster (1999–2001), an online music file sharing service created by Shawn Fanning. Although free copying and distributing of music files by

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the users led to a copyright violation lawsuit against Napster, and eventually closing of the service, it paved the way for peer-to-peer file-distribution models that came later. Gnutella had its first release in March 2000. It was followed by FastTrack (used by the Kazaa), BitTorrent, WinMX, and GNUnet in March, April, May, and November of 2001 respectively.

Internet users that are ready to share their resources become peers and form a network. When a peer in the network has a file (for example, an audio or video file) to share, it makes it available to the rest of the peers. An interested peer can connect itself to the computer where the file is stored and download it. After a peer downloads a file, it can make it available for other peers to download. As more peers join and download that file, more copies of the file become available to the group. Since lists of peers may grow and shrink, the question is how the paradigm keeps track of loyal peers and the location of the files. To answer this question, we first need to divide the P2P networks into two categories: centralized and decentralized.

Centralized networks

In a centralized P2P network, the directory system listing of the peers and what they offer uses the client–server paradigm, but the storing and downloading of the files are done using the peer-to-peer paradigm. For this reason, a centralized P2P network is sometimes referred to as a hybrid P2P network. Napster was an example of a centralized P2P. In this type of network, a peer first registers itself with a central server. The peer then provides its IP address and a **list** of files it has to share. To avoid system collapse, Napster used several servers for this purpose, but we show only one in Figure 6.18.

A peer, looking for a particular file, sends a query to a central server. The server searches its directory and responds with the IP addresses of nodes that have a copy of the file. The peer contacts one of the nodes and downloads the file. The directory is constantly updated as nodes join or leave the peer.

Centralized networks make the maintenance of the directory simple but have several drawbacks. Accessing the directory can generate huge traffic and slow down the system. The central servers are vulnerable to attack, and if all of them fail, the whole system goes down. The central component of the system was ultimately responsible for Napster's losing the copyright lawsuit and its closure in July 2001. Roxio brought back the New Napster in 2003; Napster version 2 is now a legal, pay-for-music site.

Decentralized network

A decentralized P2P network does not depend on a centralized directory system. In this model, peers arrange themselves into an *overlay network*, which is a logical network made on top of the physical network. Depending on how the nodes in the overlay network are linked, a decentralized P2P network is classified as either unstructured or structured.

In an unstructured P2P network, the nodes are linked randomly. A search in an unstructured P2P is not very efficient because a query to find a file must be flooded through the network, which produces significant traffic and still the query may not be resolved. Two examples of this type of network are Gnutella and Freenet. We next discuss the Gnutella network as an example.

A structured network uses a predefined set of rules to link nodes so that a query can be effectively and efficiently resolved. The most common technique used for this purpose is the *Distributed Hash Table (DHT)*. DHT is used in many applications including Distributed Data Structure (DDS), Content Distributed Systems (CDS), Domain Name

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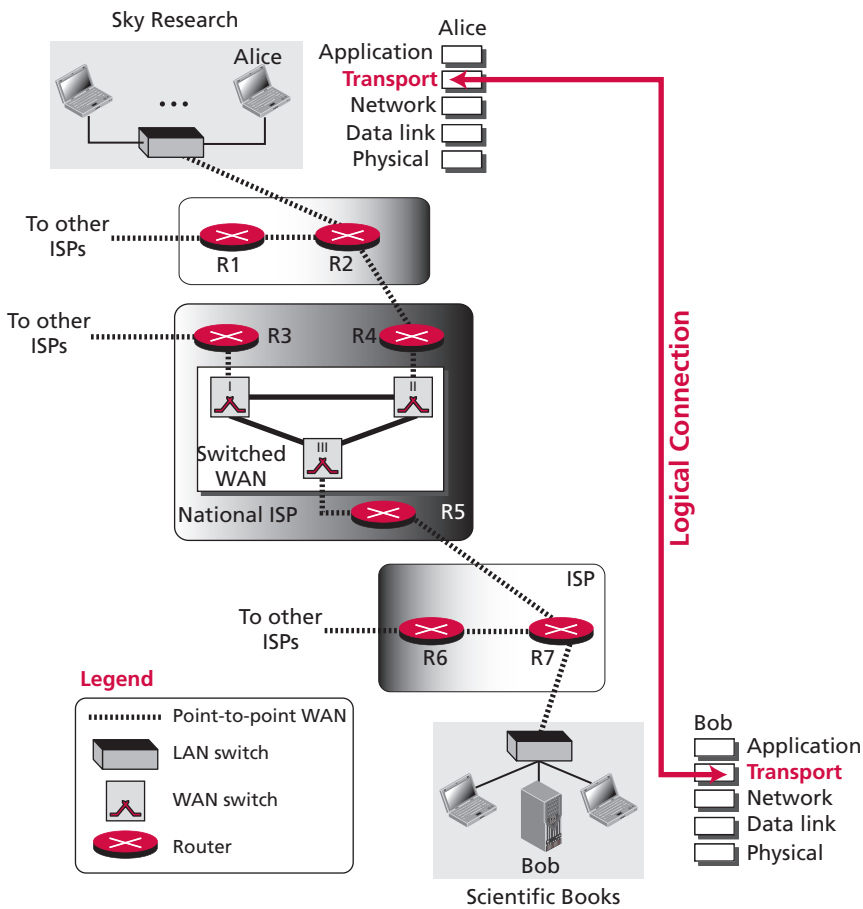
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System (DNS), and P2P file sharing. One popular P2P file sharing protocol that uses the DHT is BitTorrent. We discuss DHT independently in the next section as a technique that can be used both in structured P2P networks and in other systems.

6.3 TRANSPORT LAYER

The **transport layer** in the TCP/IP suite is located between the application layer and the network layer. It provides services to the application layer and receives services from the network layer. The transport layer acts as a liaison between a client program and a server program, a process-to-process connection. The transport layer is the heart of the TCP/IP protocol suite; it is the end-to-end logical vehicle for transferring data from one point to another in the Internet. Figure 6.19 shows the idea behind this logical connection.

Figure 6.19 *Logical connection at the transport layer*



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The figure shows the same scenario we used for the application layer. Alice's host in the Sky Research company creates a logical connection with Bob's host in the Scientific Books company at the transport layer. The two companies communicate at the transport layer as though there is a real connection between them. Figure 6.19 shows that only the two end systems (Alice's and Bob's computers) use the service of the transport layer; all intermediate routers use only the first three layers.

6.3.1 Transport-layer services

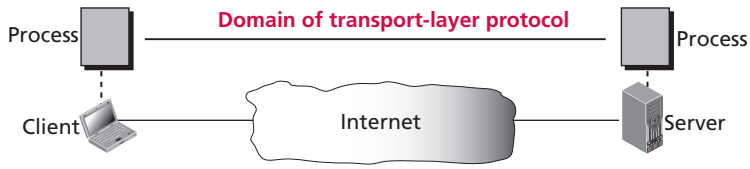
In this section, we discuss the services that can be provided by the transport layer; in the next section, we discuss several transport-layer protocols.

Process-to-process communication

The first duty of a transport-layer protocol is to provide **process-to-process communication.** A process is an application-layer entity (running program) that uses the services of the transport layer.

The network layer (discussed later) is responsible for communication at the computer level (host-to-host communication). A network-layer protocol can deliver the message only to the destination computer. However, this is an incomplete delivery. The message still needs to be handed to the correct process. This is where a transport-layer protocol takes over. A transport-layer protocol is responsible for delivery of the message to the appropriate process. Figure 6.20 shows the domains of a network layer and a transport layer.

Figure 6.20 *Network layer versus transport layer*



Addressing: port numbers

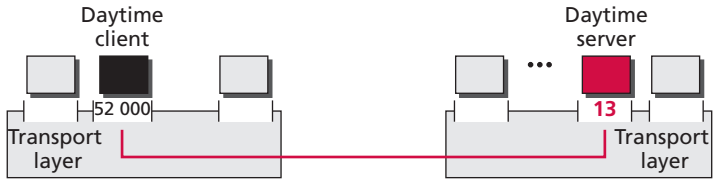
Although there are a few ways to achieve process-to-process communication, the most common is through the client–server paradigm (discussed before). A process on the local host, called a *client*, needs services from a process usually on the remote host, called a *server*. Both processes (client and server) have the same name. For example, to get the day and time from a remote machine, we need a daytime client process running on the local host and a daytime server process running on a remote machine. A remote computer can run several server programs at the same time, just as several local computers can run one or more client programs at the same time. For communication, we must define the local host, local process, remote host, and remote process. The local host and the remote host are defined using IP addresses (discussed in the next section). To define the processes, we need second identifiers, called **port numbers**. In the TCP/IP protocol suite, the port numbers are integers between 0 and 65 535 (16 bits).

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The client program defines itself with a port number, called the **ephemeral port number**. The word ephemeral means *short-lived* and is used because the life of a client is normally short. An ephemeral port number is recommended to be greater than 1023 for some client/server programs to work properly. The server process must also define itself with a port number. This port number, however, cannot be chosen randomly. TCP/IP has decided to use universal port numbers for servers; these are called **well-known port numbers**. Every client process knows the well-known port number of the corresponding server process. For example, while the daytime client process, discussed above, can use an ephemeral (temporary) port number, 52 000, to identify itself, the daytime server process must use the well-known (permanent) port number 13. Figure 6.21 shows this concept.

Figure 6.21 *Port numbers*



6.3.2 Transport-layer protocols

Although the Internet uses several transport-layer protocols, we discuss only two in this section: UDP and TCP.

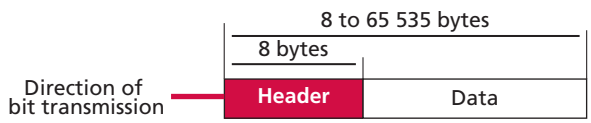
User Datagram Protocol (UDP)

The **User Datagram Protocol (UDP)** is a connectionless, unreliable transport protocol. It does not add anything to the services of network layer except for providing process-to-process communication instead of host-to-host communication. If UDP is so powerless, why would a process want to use it? With the disadvantages come some advantages. UDP is a very simple protocol using a minimum of overhead. If a process wants to send a small message and does not care much about reliability, it can use UDP. Sending a small message using UDP takes much less interaction between the sender and receiver than using TCP.

User datagrams

UDP packets, called **user datagrams**, have a fixed-size **header** of 8 bytes. Figure 6.22 shows the format of a user datagram. However, the total length needs to be less because a UDP user datagram is stored in an **IP datagram** with the total length of 65 535 bytes.

Figure 6.22 *User datagram packet format*



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Transmission Control Protocol (TCP)

Transmission Control Protocol (TCP) is a connection-oriented, reliable protocol. TCP explicitly defines connection establishment, data transfer, and connection teardown phases to provide a connection-oriented service. Connection-oriented service here means that there is a connection (relationship) between all packets (segments) belonging to the same message (coming from the application layer). TCP uses sequence numbers to define the order of the segments. The sequence number is related to the number of bytes in each segment. For example, if the message is 6000 bytes, the first segment has a sequence number 0, the second has the sequence number 2000, and the third segment has the sequence number 4000 (the process is more complicated, we try to simplify it). In this way, if a segment is lost, the receiver holds the other two until the lost one is reset by the sender.

Segments

At the transport layer, TCP groups a number of bytes together into a packet called a **segment**. TCP adds a header to each segment (for control purposes) and delivers the segment to the network layer for transmission. The segments are encapsulated in an IP datagram and transmitted as shown in Figure 6.23.

Figure 6.23 *TCP segments*



6.4 NETWORK LAYER

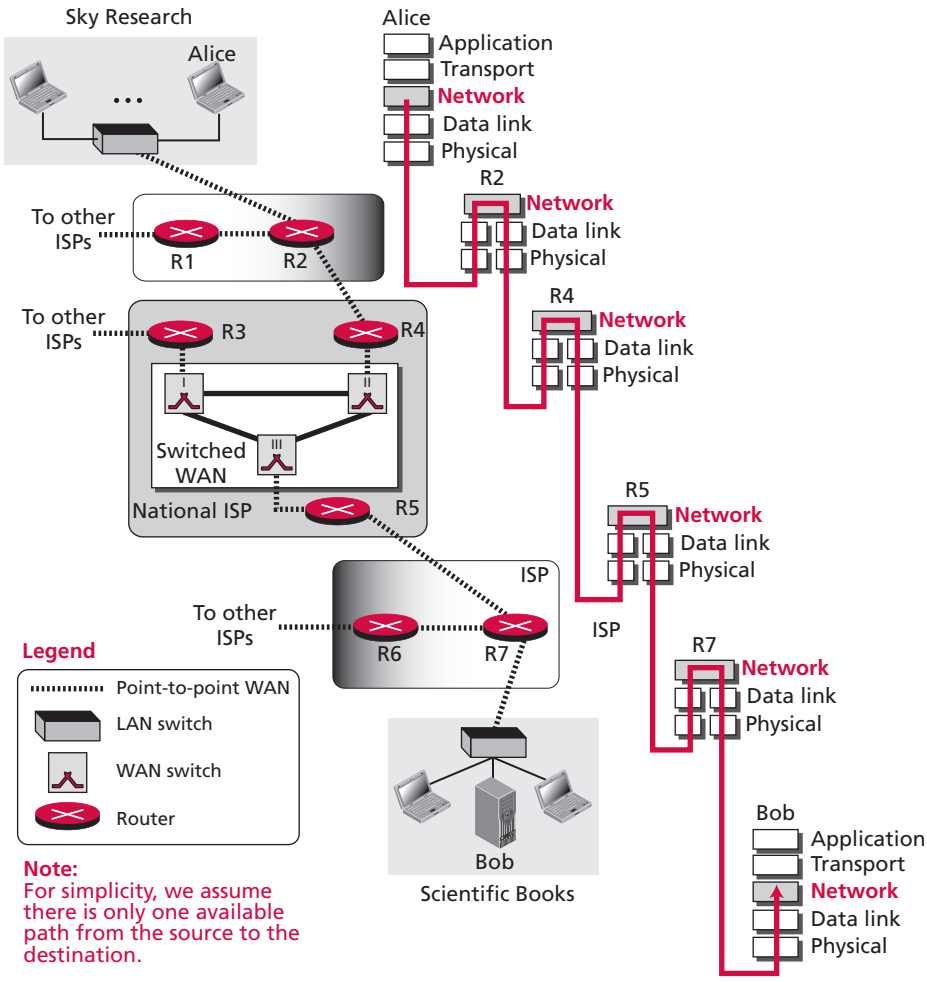
The network layer in the TCP/IP protocol suite is responsible for the host-to-host delivery of messages. Figure 6.24 shows the communication between Alice and Bob at the network layer assuming that there is only one path from Alice's computer to Bob's computer. This is the same scenario we used in the last two sections to show the communication at the application and the transport layers, respectively.

As the figure shows, the network layer is involved at the source host, destination host, and all routers in the path (R2, R4, R5, and R7). At the source host (Alice), the network layer accepts a packet from a transport layer, encapsulates the packet in a **datagram**, and delivers the packet to the data-link layer. At the destination host (Bob), the datagram is decapsulated, the packet is extracted and delivered to the corresponding transport layer. Although the source and destination hosts are involved in all five layers of the TCP/IP suite, the routers use three layers if they are routing packets only; however, they may need the transport and application layers for control purposes. A router in the path is normally shown with two data-link layers and two physical layers, because it receives a packet from one network and delivers it to another network.

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Figure 6.24 *Communication at the network layer*



6.4.1 Services Provided by network layer

The network layer is located under the transport layer; this means that the network layer provides service to the transport layer. We discuss some aspects of this service below.

Packetizing

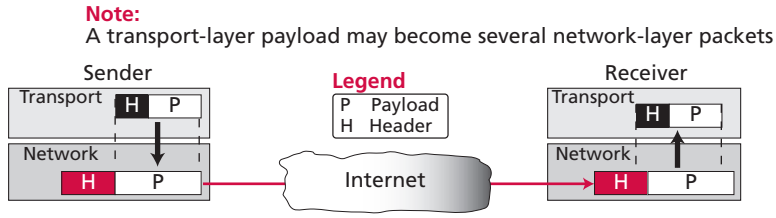
The first duty of the network layer is definitely **packetizing**: encapsulating the payload (data received from the upper layer) in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination. In other words, one duty of the network layer is to carry a payload from the source to the destination without changing it or using it. The network layer is doing the service of a carrier such as the post

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office, which is responsible for delivery of packages from a sender to a receiver without changing or using the contents. This is done in three steps as shown in Figure 6.25.

Figure 6.25 *Packetizing at the network layer*



1. The source network-layer protocol receives a packet from the transport-layer protocol, adds a header that contains the source and destination addresses and some other information that is required by the network-layer protocol.
2. The network layer protocol then logically delivers the packet to the network-layer protocol at the destination.
3. The destination host receives the network-layer packet, decapsulates the payload and delivers to the upper-layer protocol.

If the packet is fragmented at the source or at routers along the path, the network layer is responsible for waiting until all fragments arrive, reassembling them, and delivering them to the upper-layer protocol.

A transport-layer payload may be encapsulated in several network-layer packets.

Packet delivery

Packet delivery at the network layer is unreliable and connectionless. We briefly discuss these two concepts next.

Unreliable delivery

The delivery of packets at the network layer is *unreliable*. This means that the packets can be corrupted, lost, duplicated. In other words, the network layer provides a best-effort delivery, but there is no guarantee that a packet will reach the destination as we expect. This is the same service we receive from the post office when we mail a regular letter. The reason in both cases is the cost. If we need a guarantee from the post office, the cost is higher (registered mail for example). If we need a guarantee from the network layer, the delivery of packets will be delayed. Each packet needs to be checked at each router and destination and resent if corrupted. Checking the lost packets is even more costly. Does it mean that messages we send by the Internet are not reliable? The answer is that if we want to guarantee that the messages are not corrupted, we need to use the TCP protocol at the transport layer. If a payload at the transport layer is corrupted (because of unreliable

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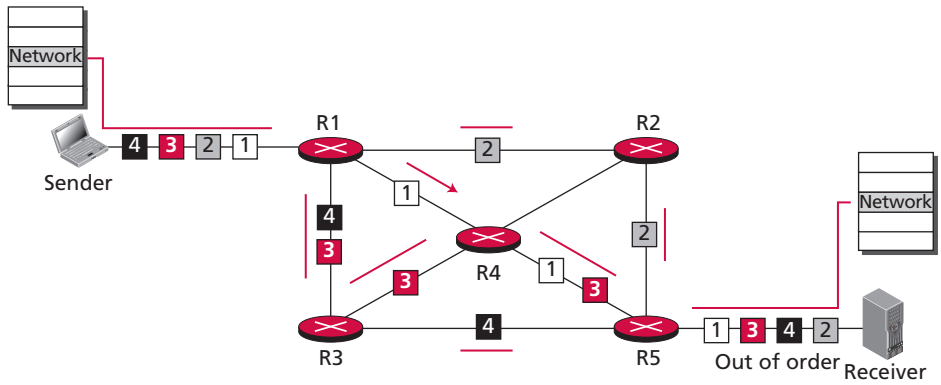
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delivery at the data-link layer), TCP drops the packet and requests resending of the data as we discussed in the previous section.

Connectionless delivery

The delivery at the network layer is also connectionless, but the word *connectionless* here does not mean that there is no physical connection between the sender and receiver. It means that the network layer treats each packet independently (like the way the post office does with the letters). In other words, there is no relationship between packets belonging to the same transport-layer payload. If a transport-layer packet results in four network-layer packets, there is no guarantee that the packets arrive in the same order as sent because each packet may follow a different path to reach the destination. Figure 6.26 shows the reason for this problem.

Figure 6.26 *Packets travelling different paths*



A transport-layer packet is divided into four network-layer packets. They are sent in order (1, 2, 3, 4), but they are received out of order (2, 4, 3, 1). The transport layer at the destination is responsible for holding packets until all of them arrive before putting them in order and delivering them to the application layer.

Routing

Another duty of the network layer, which is as important as the others, is routing. The network layer is responsible for routing the packet from its source to the destination. A physical network is a combination of networks (LANs and WANs) and routers that connect them. This means that there is more than one route from the source to the destination. The network layer is responsible for finding the *best* one among these possible routes. The network layer needs to have some specific strategies for defining the best route. In the Internet today, this is done by running some *routing protocols* to help the routers coordinate their knowledge about the neighborhood and to come up with consistent tables to be used when a packet arrives.

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6.4.2 Network-layer protocols

Although there are several protocols at the network layer, the main protocol is called the **Internet Protocol (IP)**. Other protocols are auxiliary protocols that help IP. Today, two versions of IP protocol are in use: IPv4 and IPv6. We discuss each in the next two sections.

Internet Protocol Version 4 (IPv4)

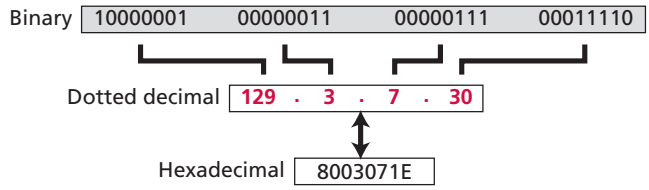
Today most systems are using the Internet Protocol Version 4 (IPv4), but this will be changed in future because of smaller address space and packet format of this protocol (among other reasons).

IPv4 addressing

The identifier used in the IPv4 layer of the TCP/IP protocol suite to identify the connection of each device to the Internet is called the **Internet address** or IP address. An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a host or a router to the Internet. The IP address is the address of the connection, not the host or the router, because if the device is moved to another network, the IP address may be changed. IPv4 addresses are unique in the sense that each address defines one, and only one, connection to the Internet. If a device, such as a router, has several connections to the Internet, via several networks, it has several IPv4 addresses. IPv4 addresses are universal in the sense that the addressing system must be accepted by any host that wants to be connected to the Internet.

There are three common notations to show an IPv4 address: binary notation (base 2), **dotted-decimal notation** (base 256), and hexadecimal notation (base 16). In *binary notation*, an IPv4 address is displayed as 32 bits. To make the address more readable, one or more spaces are usually inserted between each octet (8 bits). Each octet is often referred to as a byte. To make the IPv4 address more compact and easier to read, it is usually written in decimal form with a decimal point (dot) separating the bytes. This format is referred to as *dotted-decimal notation*. Note that because each byte (octet) is only 8 bits, each number in the dotted-decimal notation is between 0 and 255. We sometimes see an IPv4 address in hexadecimal notation. Each hexadecimal digit is equivalent to four bits. This means that a 32-bit address has 8 hexadecimal digits. This notation is often used in network programming. Figure 6.27 shows an IP address in the three discussed notations.

Figure 6.27 *Address notation*



In any communication network that involves delivery, such as a telephone network or a postal network, the addressing system is hierarchical. In a postal network, the postal

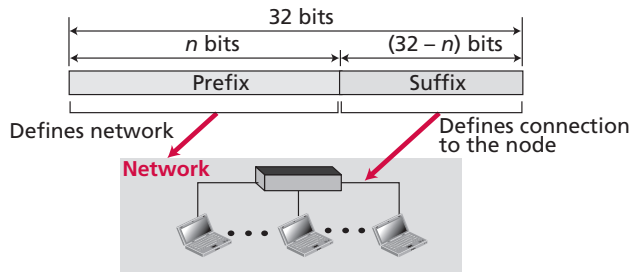
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address (mailing address) includes the country, state, city, street, house number, and the name of the mail recipient. Similarly, a telephone number is divided into the country code, area code, local exchange, and the connection.

A 32-bit IPv4 address is also hierarchical, but divided only into two parts. The first part of the address, called the *prefix*, defines the network; the second part of the address, called the *suffix*, defines the node (connection of a device to the Internet). Figure 6.28 shows the prefix and suffix of a 32-bit IPv4 address. The prefix length is *n* bits and the suffix length is (32 – *n*) bits. The prefix and suffix lengths depend on the site of the network (organization).

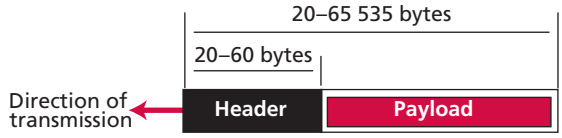
Figure 6.28 *Hierarchy in IPv4 addressing*



IPv4 datagram

Packets used by the IP are called *datagrams*. Figure 6.29 shows the IPv4 datagram format. A datagram is a variable-length packet consisting of two parts: header and payload (data). The header is 20 to 60 bytes in length and contains information essential to routing and delivery. Note that a byte is 8 bits.

Figure 6.29 *IPv4 datagram*



Internet Protocol Version 6 (IPv6)

Some shortcomings of IPv4 such as address depletion prompted a new version of IP protocol in the early 1990s. The new version, which is called **Internet Protocol version 6 (IPv6)** or **IP new generation (IPng)** was a proposal to augment the address space of IPv4 and at the same time redesign the format of the IP packet and revise some auxiliary protocols. It is

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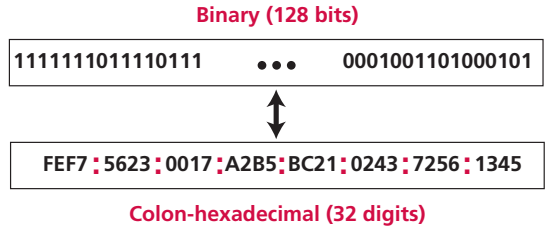
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interesting to know that IPv5 was a proposal that never materialized. The following shows the main changes in the IPv6 protocol.

IPv6 addressing

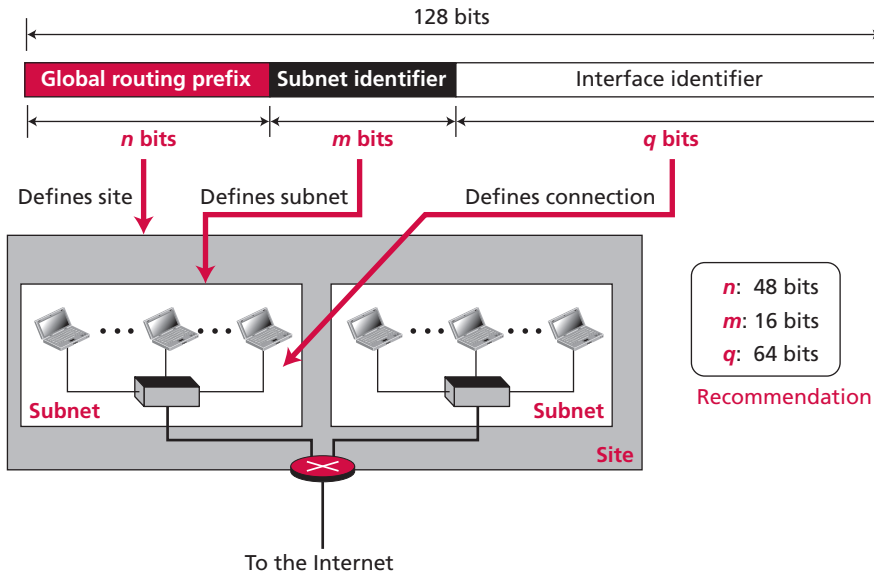
To prevent the address depletion, IPv6 uses 128 bits to define any device connected to the Internet. An address is represented as either binary or colon-hexadecimal form. The first form is used to store an address in the computer; the second form is used by humans. Figure 6.30 shows the two formats.

Figure 6.30 *IPv6 address notation*



The address in IPv6 actually defines three levels of hierarchy: site (organization), subnetwork, and connection to the host as shown in Figure 6.31.

Figure 6.31 *Hierarchy in IPv6 addressing*



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IPv6 datagram

Figure 6.32 shows the IPv6 datagram format. A datagram in this version is also a variable-length packet consisting of two parts: header and payload (data). The header is 40 bytes. However, some extension headers are considered part of the payload in this version.

Figure 6.32 *IPv6 datagram*



6.5 DATA-LINK LAYER

The TCP/IP suite does not define any protocol in the **data-link layer**. This layer is the territory of networks that when connected make up the Internet. These networks, wired or wireless, receive services and provide services to the network layer. This may give us a clue that there are several standard protocols in the market today.

In the previous section, we learned that communication at the network layer is host-to-host. The Internet, however, is a combination of networks glued together by connecting devices (routers or switches). If a datagram is to travel from a host to another host, it needs to pass through these networks.

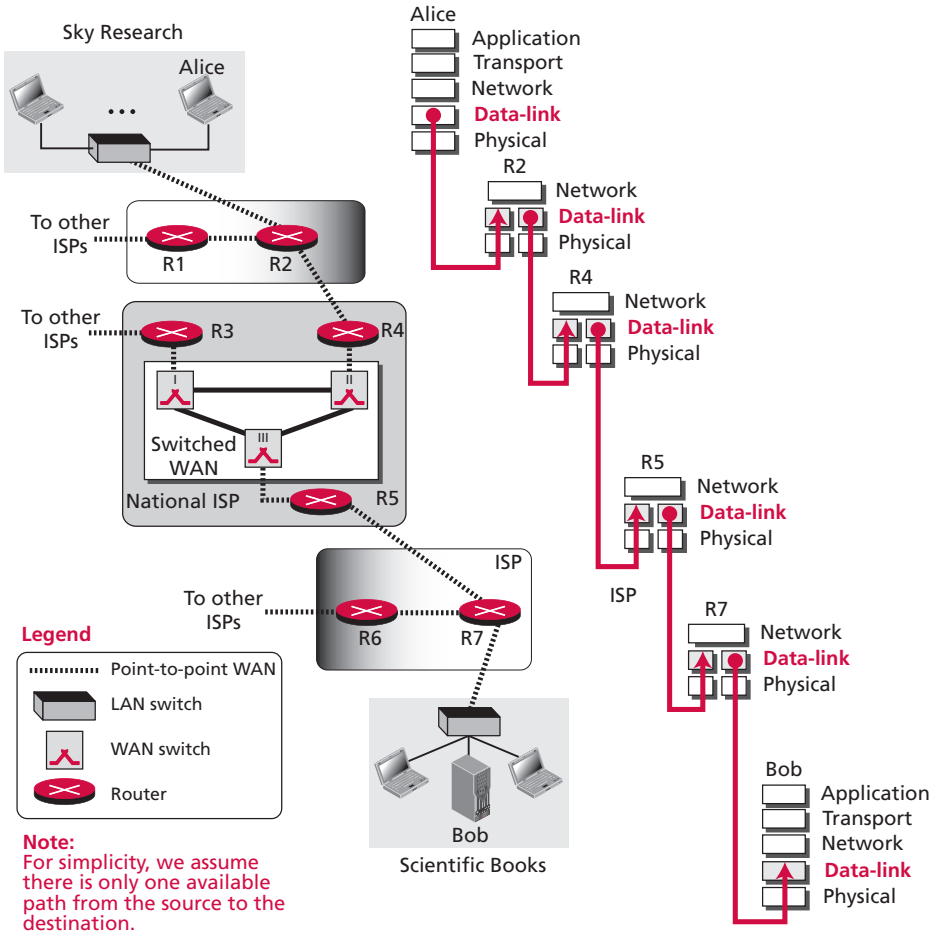
Figure 6.33 shows communication between Alice and Bob, using the same scenario we followed in the last three sections. Communication at the data-link layer, however, is made up of five separate logical connections between the data-link layers in the path.

Only one data-link layer is involved at the source or the destination, but two data-link layers are involved at each router. The reason is that Alice's and Bob's computers are each connected to a single network; each router, however, takes input from one network and sends output to another network.

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Figure 6.33 *Communication at the data-link layer*



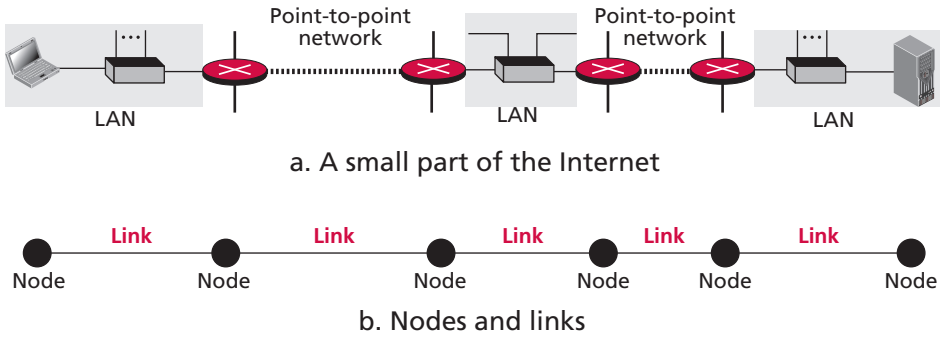
6.5.1 Nodes and links

Although communication at the application, transport, and network layers is end-to-end, communication at the data-link layer is **node-to-node**. Data units from one point in the Internet need to pass through many networks (LANs and WANs) to reach another point. These LANs and WANs are connected by routers. It is customary to refer to the two end hosts and the routers as **nodes** and the networks in between as **links.** Figure 6.34 is a simple representation of links and nodes when the path of the data unit is only six nodes.

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Figure 6.34 *Nodes and links*



The link that connects the nodes are either local area networks (LANs) or Wide Area Networks (WANs).

6.5.2 Local area networks (LANs)

In the beginning of this chapter, we learned that a local area network (LAN) is a computer network that is designed for a limited geographic area such as a building or a campus. Although a LAN can be used as an isolated network to connect computers in an organization for the sole purpose of sharing resources, most LANs today are also linked to a wide area network (WAN) or the Internet.

LANs can be wired or wireless networks. In the first group, the stations in the LANs are connected by wire; in the second group the stations are logically connected by air. We discuss each group separately.

Wired LANS: Ethernet

Although several wired LANs were invented in the past, only one has survived: the Ethernet. Maybe the reason is that Ethernet was upgraded several times according to the needs of the Internet community.

The Ethernet LAN was developed in the 1970s by Robert Metcalfe and David Boggs. Since then, it has gone through four generations: **Standard Ethernet** (10 Mbps), **Fast Ethernet** (100 Mbps), **Gigabit Ethernet** (1 Gbps), and **10 Gigabit Ethernet** (10 Gbps). The data rate, the speed in which bits are sent in each second, has been increased ten times in each generation.

Standard Ethernet

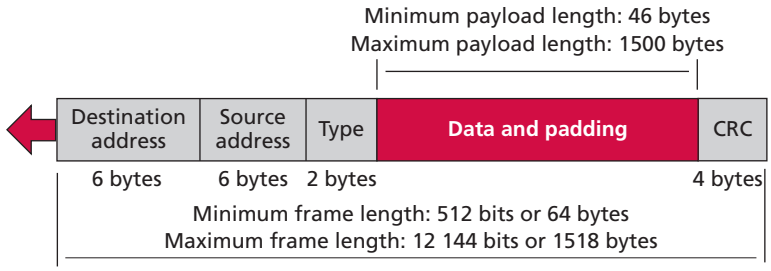
We refer to the original Ethernet technology with the data rate of 10 Mbps (ten million bits per second) as the Standard Ethernet. The data rate in this case defines the speed in which data can be sent out of the station to the LAN. In the case of the Ethernet, the speed is 10 million bits per second. The bits, however, are not sent one by one, a group of bits are packaged together and are referred to as a **frame**. A frame does not carry only data from the sender to the destination. It also carries some information such as the source

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address (48 bits), the destination address (48 bits), the type of data, the actual data, and some other control bits as a guard to help checking the integrity of data during transition. If we can think of a frame as an envelope carrying a letter from the sender to the receiver, the data is inside the envelope, but there is other information, such as addresses on the envelope. In the case of the LAN all are encapsulated in a frame. Figure 6.35 shows an Ethernet LAN and the frame format.

Figure 6.35 *Ethernet frame*



Fast Ethernet (100 Mbps)

In the 1990s, Ethernet made a big jump by increasing the **transmission rate** to 100 Mbps, and the new generation was called the Fast Ethernet. The designers of the Fast Ethernet needed to make it compatible with the Standard Ethernet. Most of the protocol such as addressing, frame format remained unchanged. By increasing the transmission rate, features of the Standard Ethernet that depend on the transmission rate had to be revised.

Gigabit Ethernet

The need for an even higher data rate resulted in the design of the **Gigabit Ethernet** Protocol (1000 Mbps). The goals of the Gigabit Ethernet were to upgrade the data rate to 1 Gbps, but keep the address length, the frame format, and the maximum and minimum frame length the same.

10-Gigabit Ethernet

In recent years, there has been another look into the Ethernet for use in metropolitan areas. The idea is to extend the technology, the data rate, and the coverage distance so that the Ethernet can be used as LAN and MAN (**metropolitan area network**). The goals of the 10-Gigabit Ethernet design can be summarized as upgrading the data rate to 10 Gbps, keeping the same frame size and format, and allowing the interconnection of LANs, MANs, and WAN possible. This data rate is possible only with fiber-optic technology at this time.

Wireless LANs

Wireless communication is one of the fastest-growing technologies. The demand for connecting devices without the use of cables is increasing everywhere. Wireless LANs can be found on college campuses, in office buildings, and in many public areas. The first difference

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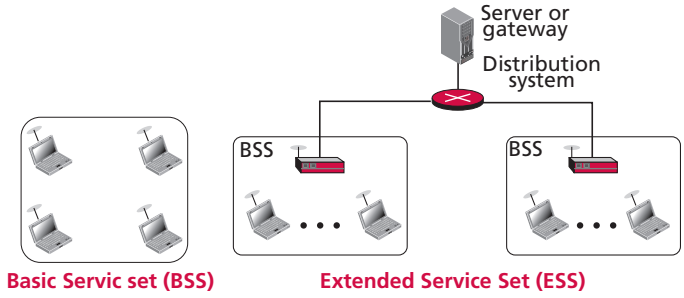
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we can see between a wired and a wireless LAN is the medium. In a wired LAN, we use wires to connect hosts. In a wireless LAN, the medium is air, the signal is generally broadcast. When hosts in a wireless LAN communicate with each other, they are sharing the same medium (multiple access). Two technologies have been instrumental in this area: Wireless Ethernet and Bluetooth.

Wireless Ethernet (WiFi)

Institute of Electrical and Electronics Engineers (IEEE) has defined the specifications for a wireless LAN, which sometimes is referred to as wireless Ethernet or *WiFi* (short for wireless fidelity). WiFi, however, is a wireless LAN that is certified by the WiFi Alliance, a global, nonprofit industry association of more than 300 member companies. The standard defines two kinds of services: the basic service set (BSS) and the extended service set (ESS). The second service uses an extra device (access point or AP) that serves as a switch for connection to other LANs or WANs. Figure 6.36 shows both services.

Figure 6.36 *BSSs and ESSs*



Bluetooth

Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, printers, and even coffee makers when they are at a short distance from each other. A Bluetooth LAN is an *ad hoc* network, which means that the network is formed spontaneously; the devices, sometimes called gadgets, find each other and make a network called a piconet. A Bluetooth LAN can even be connected to the Internet if one of the gadgets has this capability. A Bluetooth LAN, by nature, cannot be large. If there are many gadgets that try to connect, there is chaos.

Bluetooth technology has several applications. Peripheral devices such as a wireless mouse or keyboard can communicate with the computer through this technology. Monitoring devices can communicate with sensor devices in a small health care center. Home security devices can use this technology to connect different sensors to the main security controller. Conference attendees can synchronize their laptop computers at a conference.

Bluetooth was originally started as a project by the Ericsson Company. It is named after Harald Blaatand, the King of Denmark (940–981) who united Denmark and Norway. *Blaatand* translates to *Bluetooth* in English.

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6.5.3 Wide area networks (WANs)

As we discussed before, the networks connecting two nodes in the Internet can be a LAN or a WAN. As in the case of the LANs, WANs can be wired or wireless. We briefly discuss each separately.

Wired WANs

We have a variety of wired WANs in today's Ethernet. Some are point-to-point and some are switched WANs.

Point-to-point wireless WANs

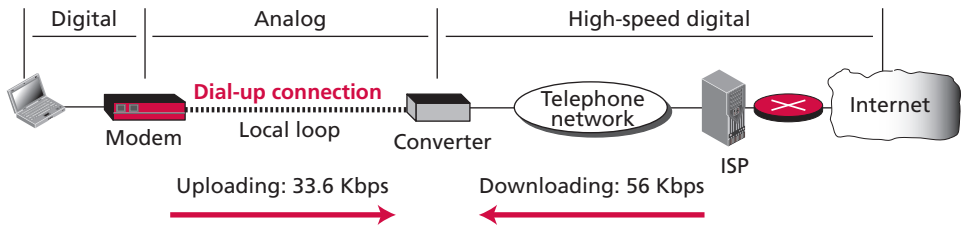
Today we can use several point-to-point wireless networks to provide what is called last-mile service to connect residents and businesses to the Internet.

Dial-up service

A dial-up network or connection uses the services provided by the telephone networks to transmit data. The telephone network had its beginnings in the late 1800s. The entire network was originally a system to transmit voice. With the advent of the computer era, the network, in the 1980s, began to carry data in addition to voice. During the last decade, the telephone network has undergone many technical changes. The need to communicate digital data resulted in the invention of the dial-up modem.

The term **modem** is a composite word that refers to the two functional entities that make up the device: a signal *mo*dulator and a signal *dem*odulator. A **modulator** creates signal from data. A **demodulator** recovers the data from the modulated signal. Figure 6.37 shows the idea behind a modem.

Figure 6.37 *Dial-up network to provide Internet access*



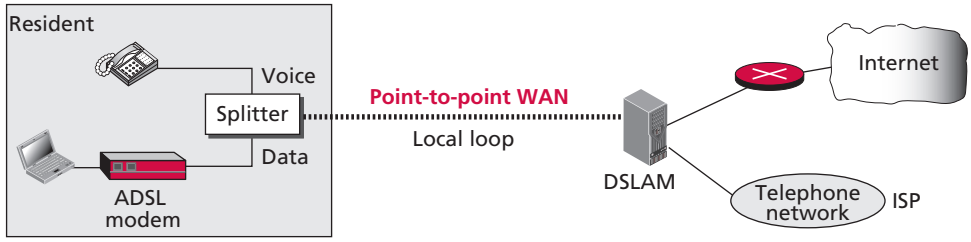
Digital subscriber line (DSL)

After traditional modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet. **Digital subscriber line (DSL)** technology is one of the most promising for supporting high-speed communication over the existing telephone. DSL technology is a set of technologies, each differing in the first letter (ADSL, VDSL, HDSL, and SDSL). The set is often referred to as *x*DSL, where *x* can be replaced by A, V, H, or S. We just discuss the first, ADSL. The first technology in the set is *asymmetric DSL (ADSL)*. ADSL provides higher speed (**bit rate**) in the downstream direction (from the Internet to the resident) than in the upstream direction (from the resident to the Internet). That is the reason it is called asymmetric (see Figure 6.38).

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Figure 6.38 *ASDL point-to-point network*



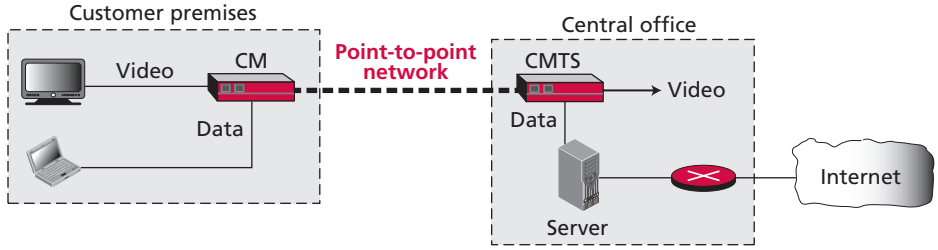
ADSL allows the subscriber to use the voice channel and the data channel at the same time. The rate for the upstream can reach 1.44 Mbps. However, the data rate is normally below 500 kbps because of the high-level noise in this channel. The downstream data rate can reach 13.4 Mbps. However, the data rate is normally below 8 Mbps because of noise in this channel. A very interesting point is that the telephone company in this case serves as the ISP, so services such as email or Internet access are provided by the telephone company itself.

Cable network

Cable networks were originally created to provide access to TV programs for those subscribers who had no reception because of natural obstructions such as mountains. Later the cable networks became popular with people who just wanted a better signal. In addition, cable networks enabled access to remote broadcasting stations via microwave connections. Cable TV also found a good market in Internet access provision, using some of the channels originally designed for video.

Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer. DSL technology provides high-data-rate connections for residential subscribers over the local loop. However, DSL uses the existing unshielded twisted-pair cable, which is very susceptible to interference. This imposes an upper limit on the data rate. A solution is the use of the cable TV network. In this section, we briefly discuss this technology. Figure 6.39 shows an example of this service.

Figure 6.39 *Cable service*



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Switched wired WANs

It is obvious that the Internet today cannot be operative with only point-to-point wired WANs that provide the last-mile connection. We need switched wired WANs to connect the backbone of the Internet. Several protocols in the past have been designed for this purpose such as SONET or ATM. However, these are complex networks, discussion of which is beyond the scope of this book.

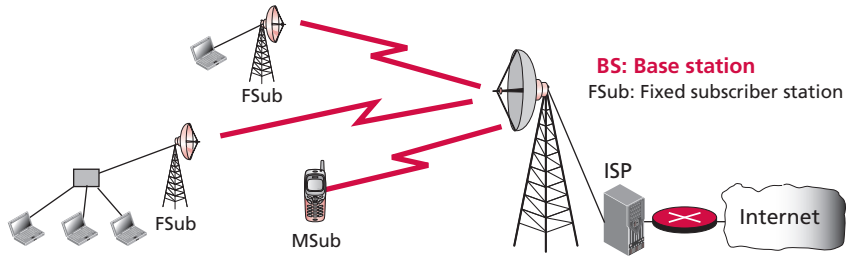
Wireless WANs

The service area of the Internet is so large today that sometimes using only wired WANs cannot provide services to each corner of the world. We definitely need wireless WANs. Several technologies have been used for this purpose as described below.

WiMax

The **Worldwide Interoperability Access (WiMax)** is the wireless version of DSL or Cable connection to the Internet. It provides two types of services (fixed WiMax) to connect the main station to fixed station or to mobile stations such as cellular phones. Figure 6.40 shows both type of connections.

Figure 6.40 *WiMax*



Cellular telephony network

Another wireless WAN today is the **cellular telephony** which was originally designed for voice communication, but it is also used today for Internet communication. We all know that the cellular network divides the earth into cells. The mobile stations communicate with the fixed antenna in the cell that they are inside at each moment. When the user moves to another cell, the communication is between the mobile device and the new antenna.

Satellite networks

A *satellite network* is a combination of nodes, some of which are satellites, that provides communication from one point on the Earth to another. A node in the network can be a satellite, an Earth station, or an end-user terminal or telephone.

Satellite networks are like cellular networks in that they divide the planet into cells. Satellites can provide transmission capability to and from any location on Earth, no matter how remote. This advantage makes high-quality communication available to less well-developed parts of the world without requiring a huge investment in ground-based infrastructure.

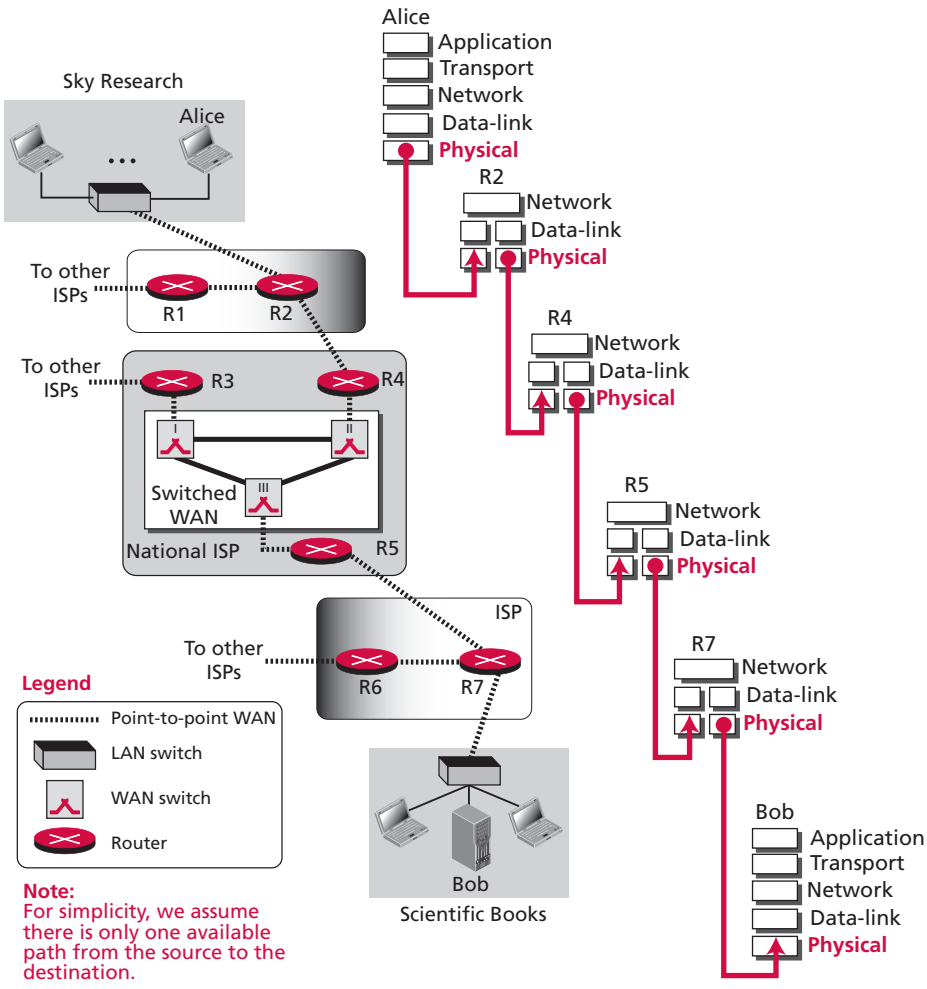
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6.6 PHYSICAL LAYER

Our discussion of the TCP/IP protocol suite would be incomplete without the discussion of the physical layer. The role of the physical layer is to transfer the bits received from the data-link layer and convert them to electromagnetic signals for transmission. After the bits are converted to signals, the signals are delivered to the transmission media, which are the subject of our discussion in the next section. Figure 6.41 uses the same scenario we showed in four earlier sections, but the communication is now at the physical layer.

Figure 6.41 *Communication at the physical layer*



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6.6.1 Data and signals

At the physical layer, the communication is node-to-node, but the nodes exchange electromagnetic signals.

One of the major functions of the physical layer is to route bits between nodes. However, bits, as the representation of two possible values stored in the memory of a node (host, router, or switch), cannot be sent directly to the transmission medium (wire or air); the bits need to be changed to signals before transmission. So the main duty of the physical layer is to efficiently convert these bits into electromagnetic signals. We first need to understand the nature of the data and then the types of signals to see how we can do this conversion efficiently.

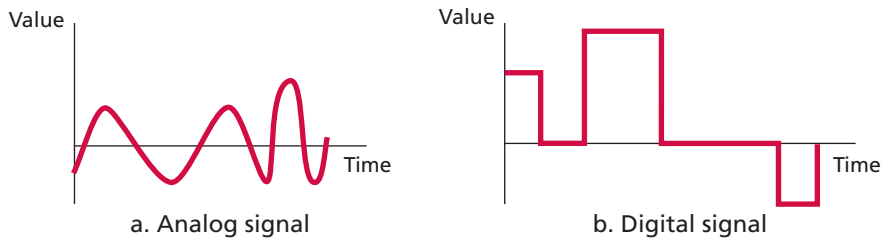
Analog and Digital

Data can be analog or digital. The term **analog data** refers to information that is continuous. Analog data, such as the sounds made by a human voice, take on continuous values. When someone speaks, an analog wave is created in the air. This can be captured by a microphone and converted to an analog signal or sampled and converted to a digital signal.

Digital data take on discrete values. For example, data are stored in computer memory in the form of 0s and 1s. They can be converted to a digital signal or modulated into an analog signal for transmission across a medium.

Like the data they represent, *signals* can be either analog or digital. An **analog signal** has infinitely many levels of intensity over a period of time. As the wave moves from value *A* to value *B*, it passes through and includes an infinite number of values along its path. A **digital signal**, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0. The simplest way to show signals is by plotting them on a pair of perpendicular axes. The vertical axis represents the value or strength of a signal. The horizontal axis represents time. Figure 6.42 illustrates an analog signal and a digital signal.

Figure 6.42 *Comparison of analog and digital signals*



6.6.2 Digital transmission

A computer network is designed to send information from one point to another. This information needs to be converted to either a digital signal or an analog signal for transmission. If data is digital, we need to use **digital-to-digital conversion** techniques, methods

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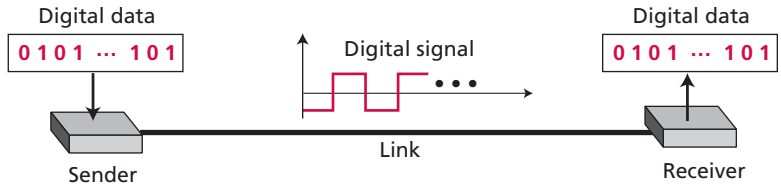
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which convert digital data to digital signals. If data is analog, we need to use **analog-to-digital conversion** techniques, methods which change an analog signal to a digital signal.

Digital–to–digital conversion

If our data is digital and we need to transmit digital signal, we can use digital-to-digital conversion to change the digital data to digital signal. Although there are many techniques for doing so, in its simplest form, a bit or group of bits is represented by a signal level as shown in Figure 6.43.

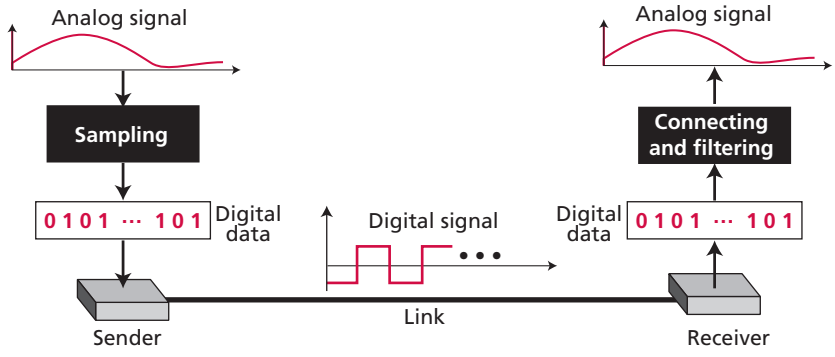
Figure 6.43 *Digital-to-digital conversion*



Analog-to-digital conversion

Sometimes we have an analog signal such as one created by a microphone or camera. The tendency today is to change an analog signal to digital data because the digital signal is less susceptible to noise. Although there are several techniques for doing so, the simplest one is to sample the analog signal to create a digital data and convert the digital data to digital signal as discussed before as shown in Figure 6.44.

Figure 6.44 *Analog-to-digital conversion*



6.6.3 Analog transmission

While digital transmission is desirable, it needs a dedicated channel; analog transmission is the only choice if we do not have a dedicated channel. For example, if we are broadcasting

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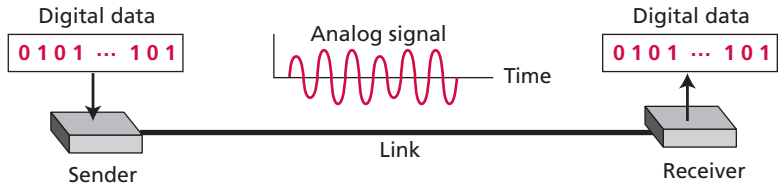
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in the air, the air belongs to everyone, so we can use only part of the channel available. Based on the data type available, we can use either digital-to-analog or analog-to analog conversion.

Digital-to-analog conversion

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data. Figure 6.45 shows the relationship between the digital information, the digital-to-analog conversion process, and the resultant analog signal.

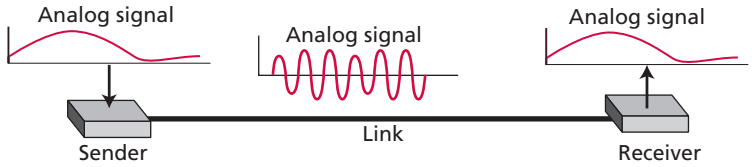
Figure 6.45 *Digital-to-analog conversion*



Analog-to-analog conversion

Analog-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data. Figure 6.46 shows the relationship between the digital information, the digital-to-analog conversion process, and the resultant analog signal.

Figure 6.46 *Analog-to-analog conversion*



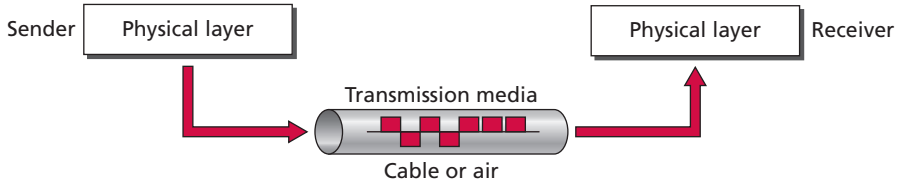
6.7 TRANSMISSION MEDIA

Electrical signals created at the physical layer need transmission media to go from one point to another. Transmission media are actually located below the physical layer and are directly controlled by the physical layer. We could say that transmission media belong to layer zero. Figure 6.47 shows the position of transmission media in relation to the physical layer.

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Figure 6.47 Transmission media and physical layer



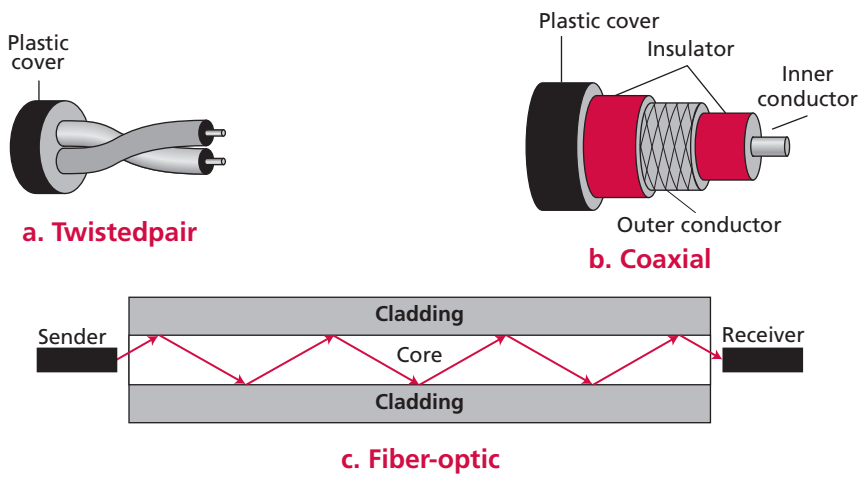
A **transmission medium** can be broadly defined as anything that can carry information from a source to a destination. For example, the transmission medium for two people having a dinner conversation is the air. The air can also be used to convey the message in a smoke signal or semaphore. For a written message, the transmission medium might be a mail carrier, a truck, or an airplane.

In telecommunications, transmission media can be divided into two broad categories: guided and unguided. Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable. Unguided medium is free space.

6.7.1 Guided media

Guided media, which are those that provide a conduit from one device to another, include **twisted-pair cable**, **coaxial cable**, and **fiber-optic cable**. Figure 6.48 shows the three types of guided media.

Figure 6.48 *Guided media*



Twisted-pair cable

A twisted-pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together. One of the wires is used to carry signals to the receiver, and

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the other is used only as a ground reference. The receiver uses the difference between the two.

In addition to the signal from the sender, interference (noise) may affect both wires and create unwanted signals. If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise sources. By twisting the pairs, a balance is maintained.

The DSL lines that are used by the telephone companies to provide high-data-rate connections are also twisted-pair cables.

Coaxial cable

Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.

Cable TV networks use coaxial cable. In the traditional cable TV network, the entire network used coaxial cable. Later, however, cable TV providers replaced most of the media with fiber-optic cable; hybrid networks use coaxial cable only at the network boundaries, near the consumer premises.

Fiber-optic cable

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. This technology uses the property of a beam of light that is refracted (comes back) when it encounters a medium of less density. Covering a glass or plastic medium by another less dense medium (called cladding) guides the light through the medium.

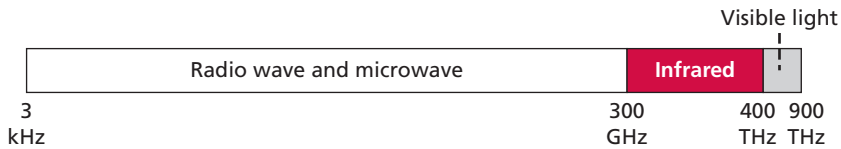
Fiber-optic cable is often found in backbone networks because its wide bandwidth is cost-effective.

6.7.2 Unguided media: wireless

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as *wireless communication*. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

Figure 6.49 shows the part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz, used for wireless communication.

Figure 6.49 *Electromagnetic spectrum for wireless communication*



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Communication today uses three different ranges of electromagnetic spectrum: radio waves, microwaves, and infrared.

Radio waves

Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves. They are used mostly for radio communication.

Microwaves

Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.

Infrared

Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room. When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

6.8 END-CHAPTER MATERIALS

6.8.1 Recommended reading

For more details about subjects discussed in this chapter, the following books are recommended.

- Forouzan, B. and Mosharrf, F. *Computer Networks: A Top-Down Approach*, New York: McGraw-Hill Education, 2012
- Forouzan, B. *Data Communication and Networking*, New York: McGraw-Hill Education, 2013
- Forouzan, B. *TCP/IP Protocol Suite*, New York: McGraw-Hill Education, 2010
- Forouzan, B. *Local Area Networks*, New York: McGraw-Hill Education, 2003
- Kurose, J. and Ross, K. *Computer Networking*, Reading, MA: Addison-Wesley, 2007

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6.8.2 Key terms

10-Gigabit Ethernet 168	internet service provider (ISP) 136
analog-to-analog conversion 177	Internetwork 134
analog-to-digital conversion 176	IP address 146
analog data 175	IP datagram 158
analog signal 175	IP new generation (IPng) 164
application layer 141	link 167
bit rate 171	local area network (LAN) 134
Bluetooth 170	Message Access Agent (MAA) 151
cellular telephony 173	message transfer agent (MTA) 151
ciphertext 138	metropolitan area network (MAN) 169
client–server paradigm 144	modem 171
coaxial cable 178	modularity 139
connecting device 134	modulator 171
connectionless protocol 182	module 139
country domain 154	name space 153
demodulator 171	network layer 142
digital-to-analog conversion 177	node 167
digital-to-digital conversion 175	packetizing 160
digital data 175	peer-to-peer (P2P) paradigm 145
digital signal 175	physical layer 141
digital subscriber line (DSL) 171	port number 142
domain name 153	protocol 137
Domain name server (DNS) 152	protocol layering 137
domain name space 153	remote login 151
dotted-decimal notation 163	router 134
electronic mail (email) 150	Secure Shell (SHH) 151
end system 134	segment 159
ephemeral port number 158	software 137
Fast Ethernet 168	source to destination delivery 182
fiber-optic cable 178	Standard Ethernet 168
File Transfer Protocol (FTP) 149	switch 134
frame 168	switched WAN 135
generic domain 153	TCP/IP protocol suite 140
Gigabit Ethernet 168	TELNET (terminal network) 151
guided media 178	Transmission Control Protocol (TCP) 159
hardware 137	transmission medium 178

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header 158	transmission rate 169
host 134	twisted-pair cable 178
host identifier 148	unguided media 179
hypertext 147	uniform resource locator (URL) 149
HyperText Markup Language (HTML) 148	user agent (UA) 151
HyperText Transfer Protocol (HTTP) 149	user datagram 158
infrared waves 180	User Datagram Protocol (UDP) 158
internet 135	web page 147
Internet 136	well-known port number 158
Internet address 163	wide area network (WAN) 134
Internet Protocol 163	Worldwide Interoperability Access (WiMAX) 173
Internet Protocol version 6 (IPv6) 164	World Wide Web (WWW) 145

6.8.3 Summary

- A network is a set of devices connected by communication links. Today when we speak of networks, we are generally referring to two primary categories: local area networks and wide area networks. The Internet today is made up of many wide and local area networks joined by connecting devices and switching stations. A protocol is a set of rules that governs communication. TCP/IP is a hierarchical protocol suite made of five layers: application, transport, network, data-link, and physical.
- Applications in the Internet are designed using either a client–server paradigm or a peer-to-peer paradigm. The World Wide Web (WWW) is a repository of information linked together from points all over the world. The HyperText Transfer Protocol (HTTP) is the main protocol used to access data on the World Wide Web (WWW). File Transfer Protocol (FTP) is a TCP/IP client–server application for copying files from one host to another. Electronic mail is one of the most common applications on the Internet. TELNET is a client–server application that allows a user to log into a remote machine, giving the user access to the remote system. The Domain Name System (DNS) is a client–server application that identifies each host on the Internet with a unique name
- The main duty of a transport-layer protocol is to provide process-to-process communication. UDP is a transport protocol that provides unreliable and connectionless service. Transmission Control Protocol (TCP) is another transport-layer protocol that provides reliable and connection-oriented service.
- The network layer supervises the handling of packets by the underlying physical networks. IPv4 is an unreliable **connectionless protocol** responsible for **source-to-destination delivery**. The identifiers used in the IP layer of the TCP/IP protocol suite

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are called the IP addresses. An IPv4 address is 32 bits long. IPv6, the latest version of the Internet Protocol, has a 128-bit address space.

- The data-link layer involves local and wide area networks (LANs and WANs). LANs and WANs can be wired or wireless. Ethernet is the most widely used wired local area network protocol. Dial-up service, DSL, and cable are mostly used for point-to-point wired WANs. Wireless LANs became formalized with wireless Ethernet. Bluetooth is a wireless LAN technology that connects devices (called gadgets) in a small area. WiMAX is a wireless access network that may replace DSL and cable in the future.
- Data must be transformed to electromagnetic signals to be transmitted. Analog data are continuous and take continuous values. Digital data have discrete states and take discrete values. Digital-to-digital conversion changes digital data to digital signal. Digital-to-analog conversion is the process of changing digital data to analog signal. Analog-to-digital conversion is the process of sampling analog data and changing it to digital signal. Analog-to-analog signal means changing analog data to analog signal.
- Transmission media lie below the physical layer. A guided medium provides a physical conduit from one device to another. Twisted-pair cable, coaxial cable, and optical fiber are the most popular types of guided media. Unguided media (free space) transport electromagnetic waves without the use of a physical conductor.

6.9 PRACTICE SET

6.9.1 Quizzes

A set of interactive quizzes for this chapter can be found on the book's website. It is strongly recommended that the student takes the quizzes to check his/her understanding of the materials before continuing with the practice set.

6.9.2 Review questions

Q6-1. What is the first principle we discussed in this chapter for protocol layering that needs to be followed to make the communication bidirectional?

Q6-2. In the TCP/IP protocol suite, what are the identical objects at the sender and the receiver sites when we think about the logical connection at the application layer?

Q6-3. A host communicates with another host using the TCP/IP protocol suite. What is the unit of data sent or received at each of the following layers?

a. application layer **b.** network layer **c.** data-link layer

Q6-4. Which of the following data units is encapsulated in a frame?

a. a user datagram **b.** a datagram **c.** a segment

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Q6-5. Which of the following data units is decapsulated from a user datagram?

a. a datagram **b.** a segment **c.** a message

Q6-6. Which of the following data units has an application-layer message plus the header from layer 4?

a. a frame **b.** a user datagram **c.** a bit

Q6-7. What are the types of addresses (identifiers) used in each of the following layers?

a. application layer **b.** network layer **c.** data-link layer

Q6-8. During the weekend, Alice often needs to access files stored on her office desktop, from her home laptop. Last week, she installed a copy of the FTP server process on her desktop at her office and a copy of the FTP client process on her laptop at home. She was disappointed when she could not access her files during the weekend. What could have gone wrong?

Q6-9. Most of the operating systems installed on personal computers come with several client processes, but normally no server processes. Explain the reason.

Q6-10. A new application is to be designed using the client–server paradigm. If only small messages need to be exchanged between the client and the server without the concern for message loss or corruption, what transport-layer protocol do you recommend?

Q6-11. Why is the routing the responsibility of the network layer? In other words, why can't the routing be done at the transport layer or the data-link layer?

Q6-12. Distinguish between communication at the network layer and communication at the data-link layer.

Q6-13. What is dial-up modem technology? List some of the common modem standards discussed in this chapter and give their data rates.

Q6-14. What is the reason that Bluetooth is normally called a wireless personal area network (WPAN) instead of a wireless local area network (WLAN)?

Q6-15. How can we find the period of a sine wave when its frequency is given?

Q6-16. Which of the following measures the value of a signal at any time?

a. amplitude **b.** frequency **c.** phase

Q6-17. Define analog transmission.

Q6-18. What is the position of the transmission media in the TCP/IP protocol suite?

Q6-19. Name the two major categories of transmission media.

Q6-20. What are the three major classes of guided media?

6.9.3 Problems

P6-1. Answer the following questions about Figure 6.5 when the communication is from Maria to Ann:

a. What is the service provided by layer 1 to layer 2 at Maria's site?

b. What is the service provided by layer 1 to layer 2 at Ann's site?

P6-2. Answer the following questions about Figure 6.5 when the communication is from Maria to Ann:

a. What is the service provided by layer 2 to layer 3 at Maria's site?

b. What is the service provided by layer 2 to layer 3 at Ann's site?

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P6-3. Assume that the number of hosts connected to the Internet at year 2010 is five hundred million. If the number of hosts increases only 20 per cent per year, what is the number of hosts in year 2020?

P6-4. Assume a system uses five protocol layers. If the application program creates a message of 100 bytes and each layer (including the fifth and the first) adds a header of ten bytes to the data unit, what is the efficiency (the ratio of application-layer bytes to the number of bytes transmitted) of the system?

P6-5. Match the following to one or more layers of the TCP/IP protocol suite:

a. route determination

b. connection to transmission media

c. providing services for the end user

P6-6. Match the following to one or more layers of the TCP/IP protocol suite:

a. creating user datagrams

b. responsibility for handling frames between adjacent nodes

c. transforming bits to electromagnetic signals

P6-7. Protocol layering can be found in many aspects of our lives such as air travelling. Imagine you make a round-trip to spend some time on vacation at a resort. You need to go through some processes at your city airport before flying. You also need to go through some processes when you arrive at the resort airport. Show the protocol layering for the round trip using some layers such as baggage checking/claiming, boarding/unboarding, takeoff/landing.

P6-8. In an internet, we change the LAN technology to a new one. Which layers in the TCP/IP protocol suite need to be changed?

P6-9. Compare the range of 16-bit addresses, 0 to 65 535, with the range of 32-bit IP addresses, 0 to 4 294 967 295. Why do we need such a large range of IP addresses, but only a relatively small range of port numbers?

P6-10. Rewrite the following IP addresses using binary notation:

a. 110.11.5.88 **b.** 12.74.16.18 **c.** 201.24.44.32

P6-11. Rewrite the following IP addresses using dotted-decimal notation:

a. **01011110 10110000 01110101 00010101**

b. **10001001 10001110 11010000 00110001**

c. **01010111 10000100 00110111 00001111**

P6-12. What is the hexadecimal equivalent of the following Ethernet address?

01011010 00010001 01010101 00011000 10101010 00001111

P6-13. A device is sending out data at the rate of 1000 bps.

a. How long does it take to send out 10 bits?

a. How long does it take to send out a character (8 bits)?

b. How long does it take to send a file of 100 000 characters?

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